

CCCW 2024

SHELXL and the SHELXL file structure

Frank Schaper – Université de Montréal

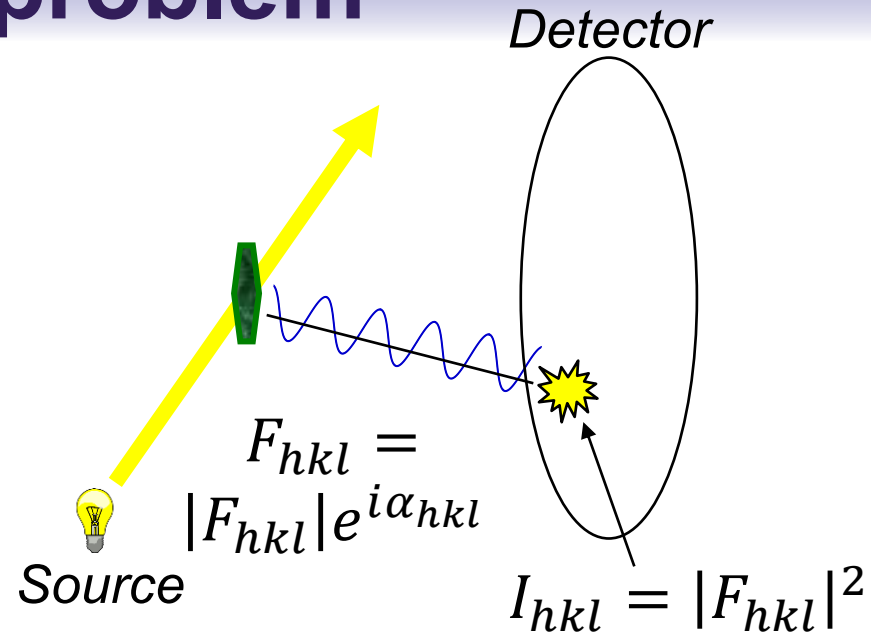
The phase problem

$$\rho(\vec{r}) = \frac{1}{V} \sum F_{hkl} e^{-2\pi i \vec{r} \cdot \vec{hkl}} d\vec{hkl}$$

$$F_{hkl} = |F_{hkl}| e^{i\alpha_{hkl}}$$

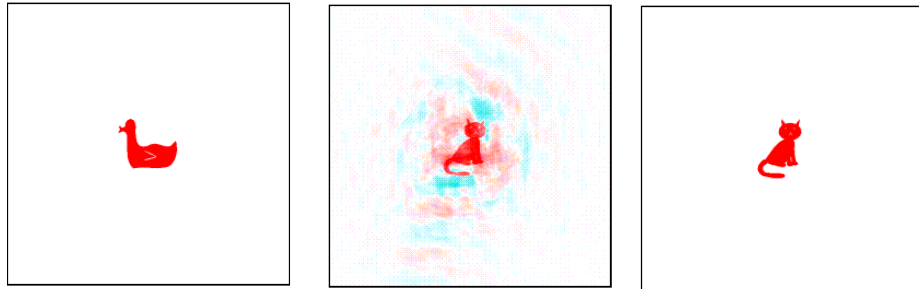
Phase of reflection hkl

Amplitude of reflection hkl
(= the wave diffracted at lattice plane hkl)



We **cannot** determine the phase of our reflection!

Are phases important ?

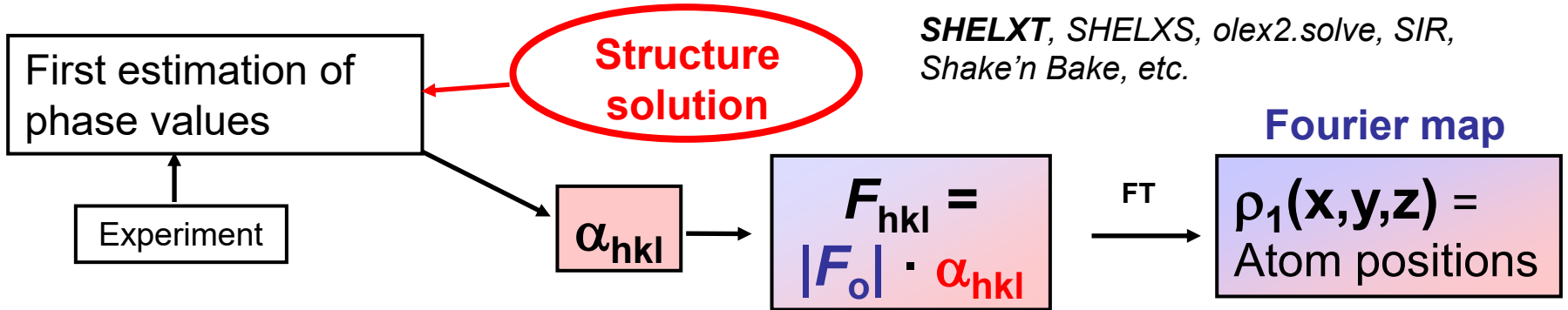


$$|F_{hk}| e^{i\alpha_{hk}} \xrightarrow{\downarrow \text{FT}} |F_{hk}| e^{i\alpha_{hk}} \xleftarrow{\uparrow \text{FT}} |F_{hk}| e^{i\alpha_{hk}}$$

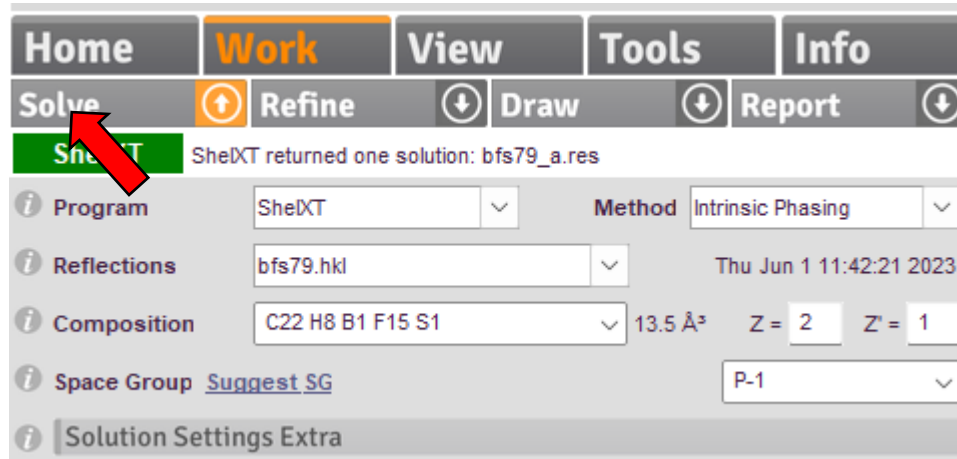


Houston, we have a
(phase) problem !

Structure solution : .ins and .res



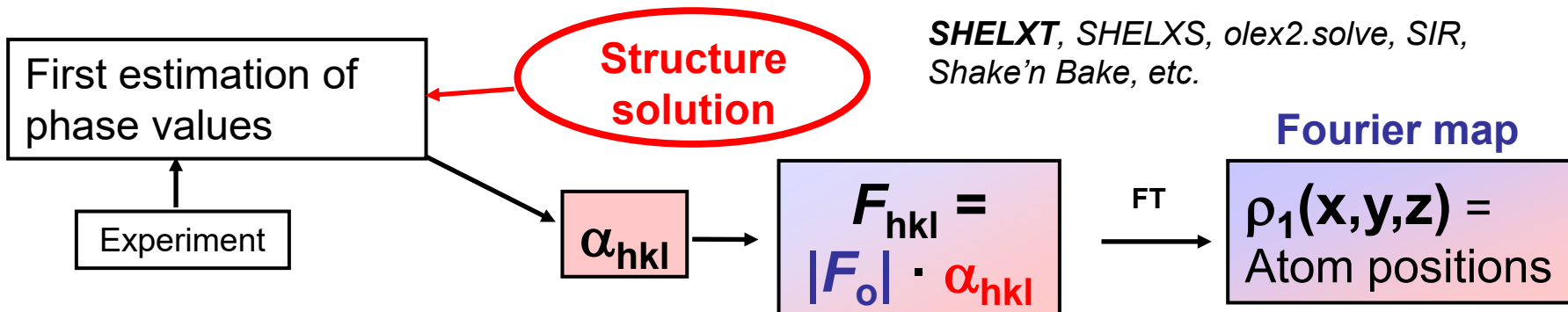
Let's solve the structure....



*If you want to
do it yourself:
open [insfile.ins](#)*

What did actually happen ?

Structure solution : .ins and .res



Let's solve the structure....

C₂₂H₈BF₁₅S

Solution

a = 9.5892(8)	α = 111.904(6)°	Z = 2
b = 10.8264(10)	β = 96.814(7)°	Z' = 1
c = 11.3959(9)	γ = 100.727(7)°	V = 1055.29(17)

d min (MoKα)	0.78	I/σ(I)	27.0	Rint	3.42%	Full 50.5°	97.5
2θ = 54.0°				m = 1.77		95% to 54.0°	
Shift	n/a	Max Peak	n/a	Min Peak	n/a	GooF	n/a

Structure **BFS79** loaded

Home **Work** **View** **Tools** **Info**

Solve **Refine** **Draw** **Report**

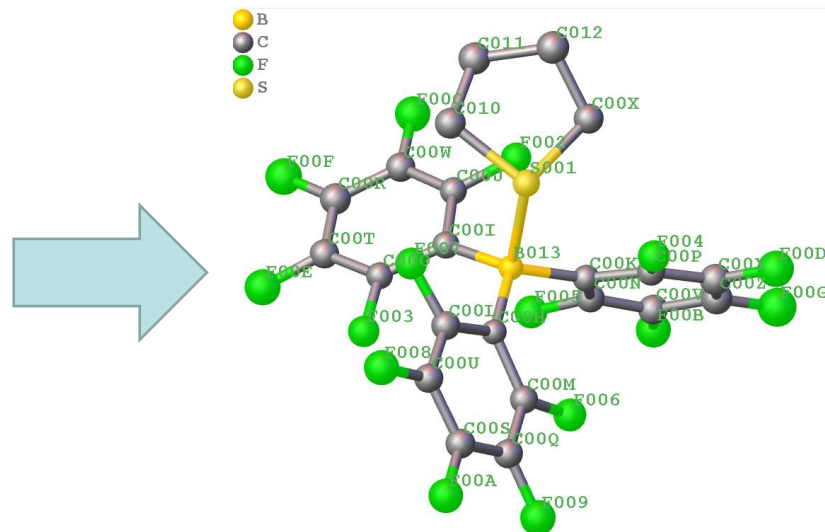
ShelXT ShelXT returned one solution: bfs79_a.res

Program: ShelXT Method: Intrinsic Phasing

Reflections: bfs79.hkl Thu May 30 09:11:18 2024

Composition: C₂₂ H₈ B₁ F₁₅ S₁ 13.5 Å³ Z = 2 Z' = 1

Space Group: [Suggest SG](#) P-1

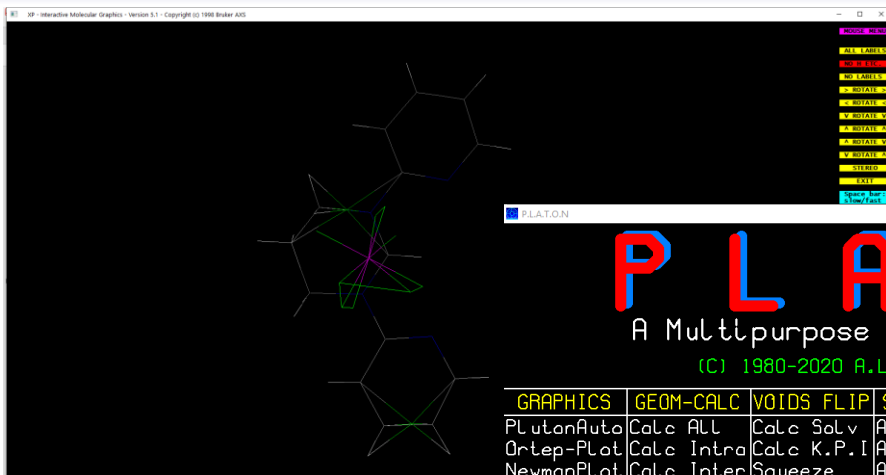


What did actually happen ?

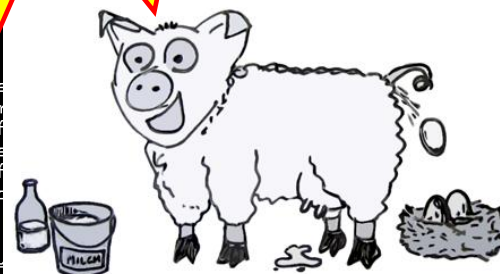
Choice of GUI



**Cool people
use OLEX2!**



**XP and Platon are for
hardcore crystallographers**



Egg-laying Woolmilkpig

WilsonPlot
PlutoNativ
Xtal Data (CIF) vlv6.clf- Set 1() : vlv6
RefL Data (LIST4) vlv6.fcf [FCF] () : vlv6
http://www.platonsft.nl/PLATON-MANUAL.pdf http://www.platonsft.nl/PLATON_HOV_T0.pdf
INPUT INSTRUCTIONS via KEYBOARD or LEFT-MOUSE-CLICKS (HELP with RIGHT CLICKS)
HEL
Reset End
Exit
MenuActive

**Nobody wants
XShell**



**Shelxle, WinGX,
Crystals, ...**

A look behind the GUI

C₂₂H₈BF₁₅S

a = 9.5892(8) **a** = 111.904(6)[°]
b = 10.8264(10) **β** = 96.814(7)[°]
c = 11.3959(9) **γ** = 100.727(7)[°]

Z = 2
Z' = 1
V = 1055.29(17)

Solution

d min (MoKα)	0.78	I/σ(I)	27.0	Rint	3.42%	Full 50.5°	97.5
2θ = 54.0°				m = 1.77		95% to 54.0°	
Shift	n/a	Max Peak	n/a	Min Peak	n/a	Goof	n/a

Structure **BFS79** loaded

Home **Work** **View** **Tools** **Info**

Solve **Refine** **Draw** **Report**

ShelXT ShelXT returned one solution: bfs79_a.res

Program ShelXT **Method** Intrinsic Phasing

Reflections bfs79.hkl Thu May 30 09:11:18 2024

Composition C22 H8 B1 F15 S1 13.5 Å³ **Z** = 2 **Z'** = 1

Space Group **Suggest SG** **P-1**

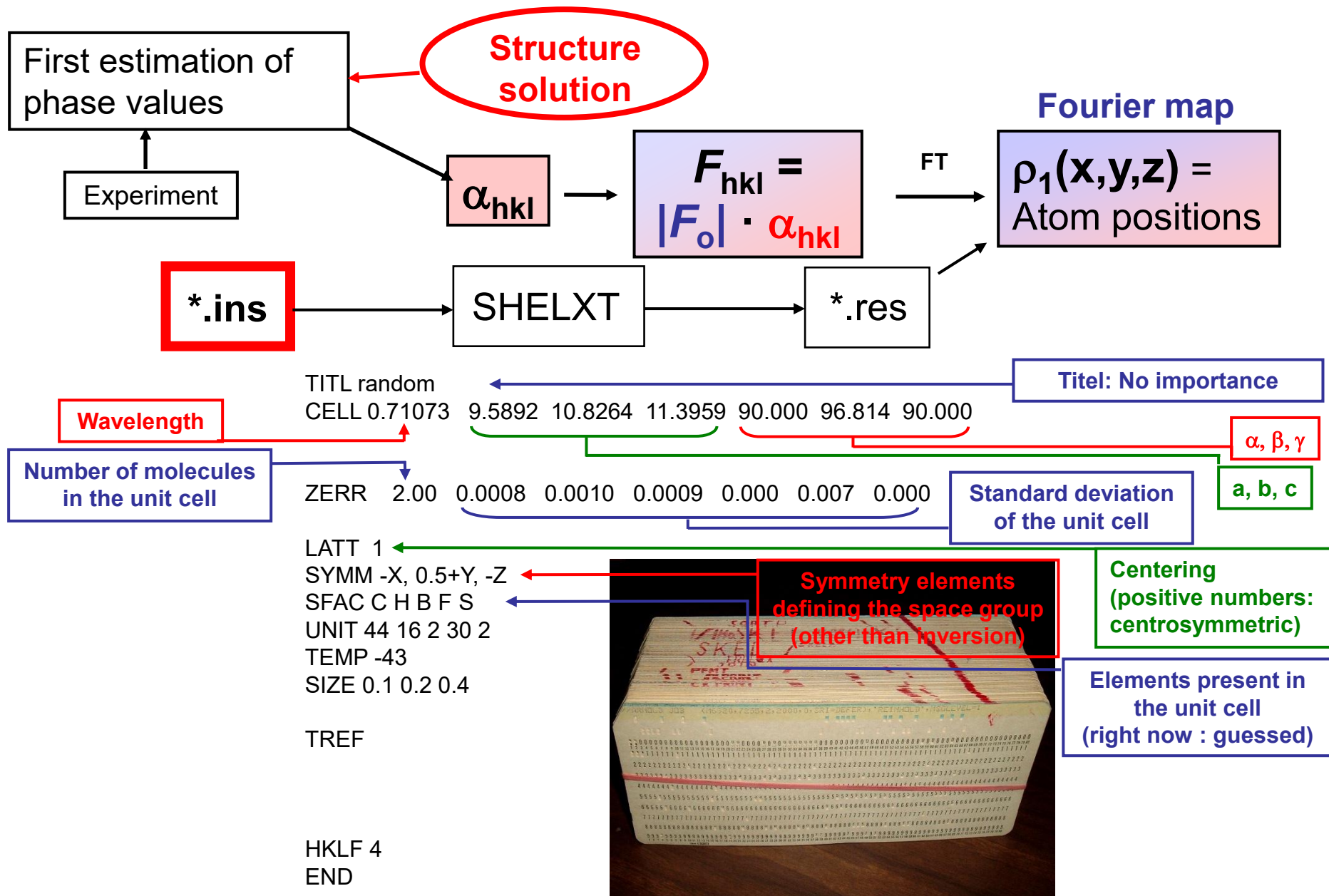
```
TITL bfs79 in P-1
CELL 0.71073 9.5892 10.8264 11.3959 111.904 96.814 100.727
ZERR 2.00 0.0008 0.0010 0.0009 0.006 0.007 0.007
LATT 1
SFAC C H B F S
UNIT 44 16 2 30 2
TEMP -43
SIZE 0.1 0.2 0.4

TREF

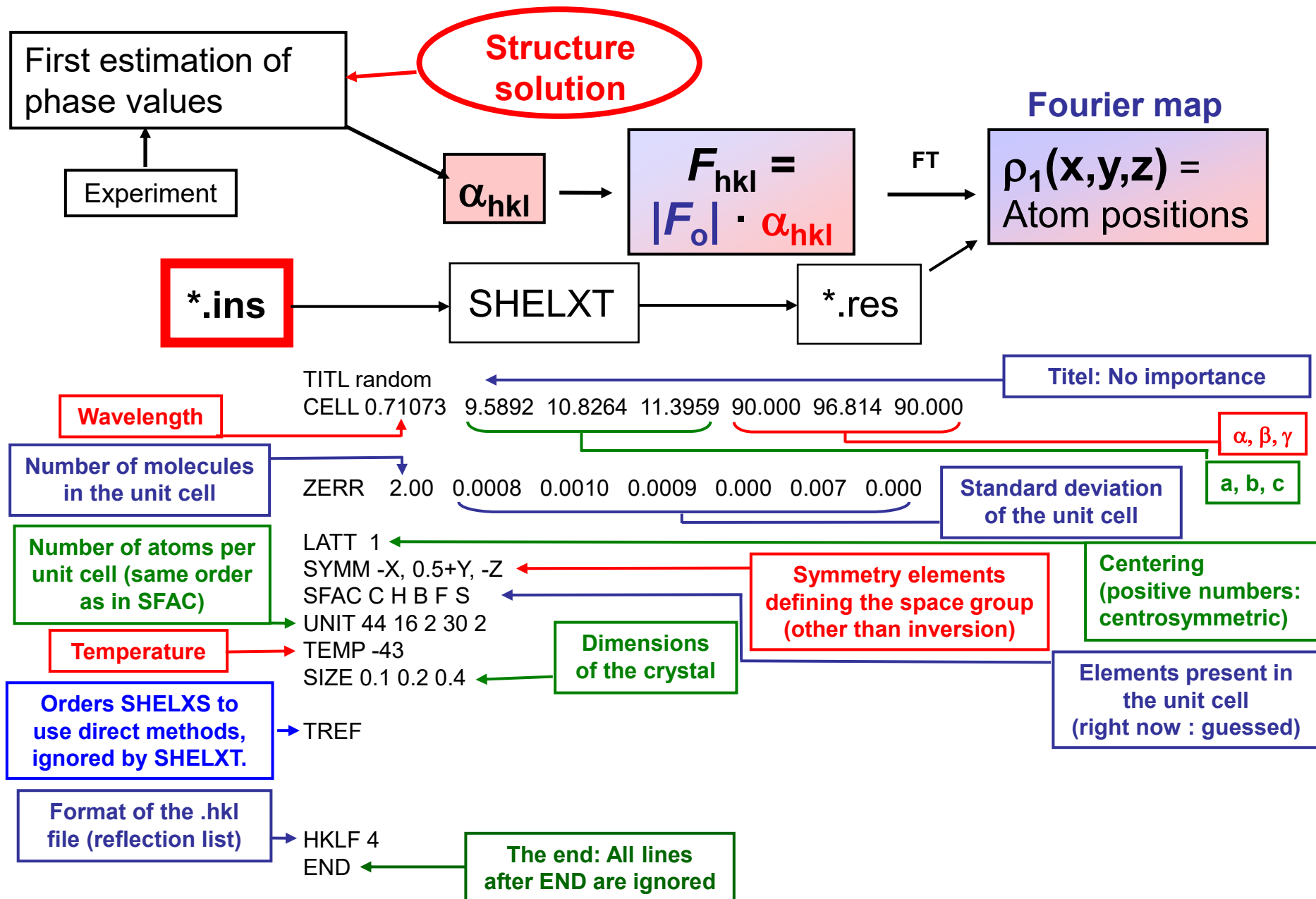
HKLF 4
END
```

****.INS

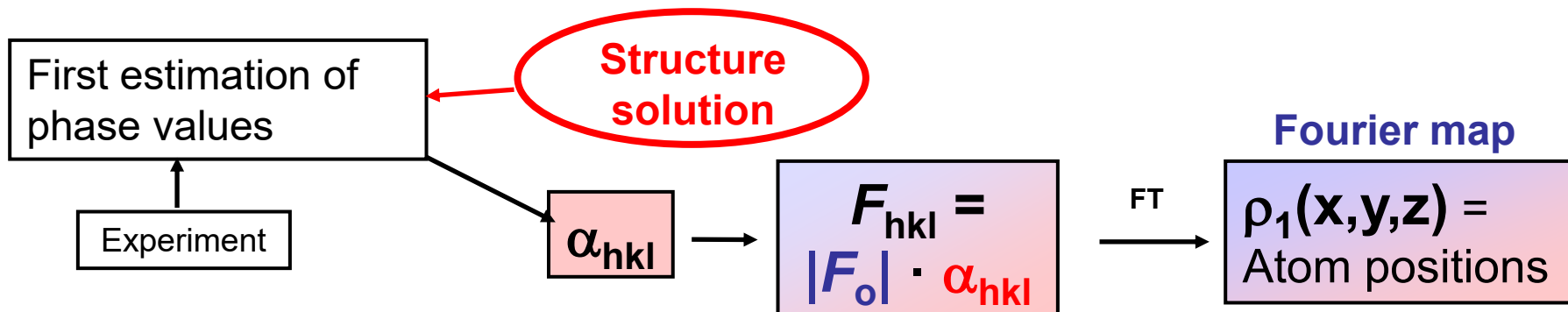
Structure solution : .ins / .res



Structure solution : .ins / .res



Structure solution : .ins and .res



TITL bfs79 in P-1

CELL 0.71073 9.5892 10.8264 11.3959 90.000 96.814 90.000

ZERR 2.00 0.0008 0.0010 0.0009 0.000 0.007 0.000

LATT 1
SYMM -X, 0.5+Y, -Z
SFAC C H B F S
UNIT 44 16 2 30 2
TEMP -43
SIZE 0.1 0.2 0.4

TREF

HKLF 4
END

These commands have **no influence** on the actual refinement. But if they are missing or wrong, you end up with wrong information in your CIF. I. e., molecular weight, absorption coefficient, etc.

You should never ever change these commands by hand!

You *might* change LATT, SYMM and HKLF... if you really, really know what you are doing.

In case we misunderstand each other, this means: **do not touch!**

A look behind the GUI

C₂₂H₈BF₁₅S

Solution

a = 9.5892(8)	α = 111.904(6)°	Z = 2
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Shift	n/a	Max Peak	n/a	Min Peak	n/a	Goof	n/a

Structure **BFS79** loaded

Home **Work** **View** **Tools** **Info**

Solve **Refine** **Draw** **Report**

ShelXT ShelXT returned one solution: bfs79_a.res

Program: **ShelXT** Method: Intrinsic Phasing

Reflections: bfs79.hkl Thu May 30 09:11:18 2024

Composition: C₂₂ H₈ B₁ F₁₅ S₁ 13.5 Å³ Z = 2 Z' = 1

Space Group: [Suggest SG](#) P-1

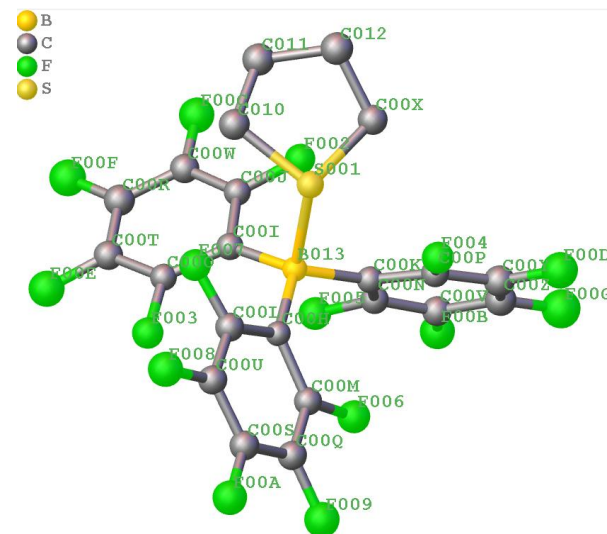
TITL bfs79 in P-1
 CELL 0.71073 9.5892 10.8264 11.3959 111.904 96.814 100.727
 ZERR 2.00 0.0008 0.0010 0.0009 0.006 0.007 0.007
 LATT 1
 SFAC C H B F S
 UNIT 44 16 2 30 2
 TEMP -43
 SIZE 0.1 0.2 0.4

******.INS**

TREF

HKLF 4
 END

SHELXT

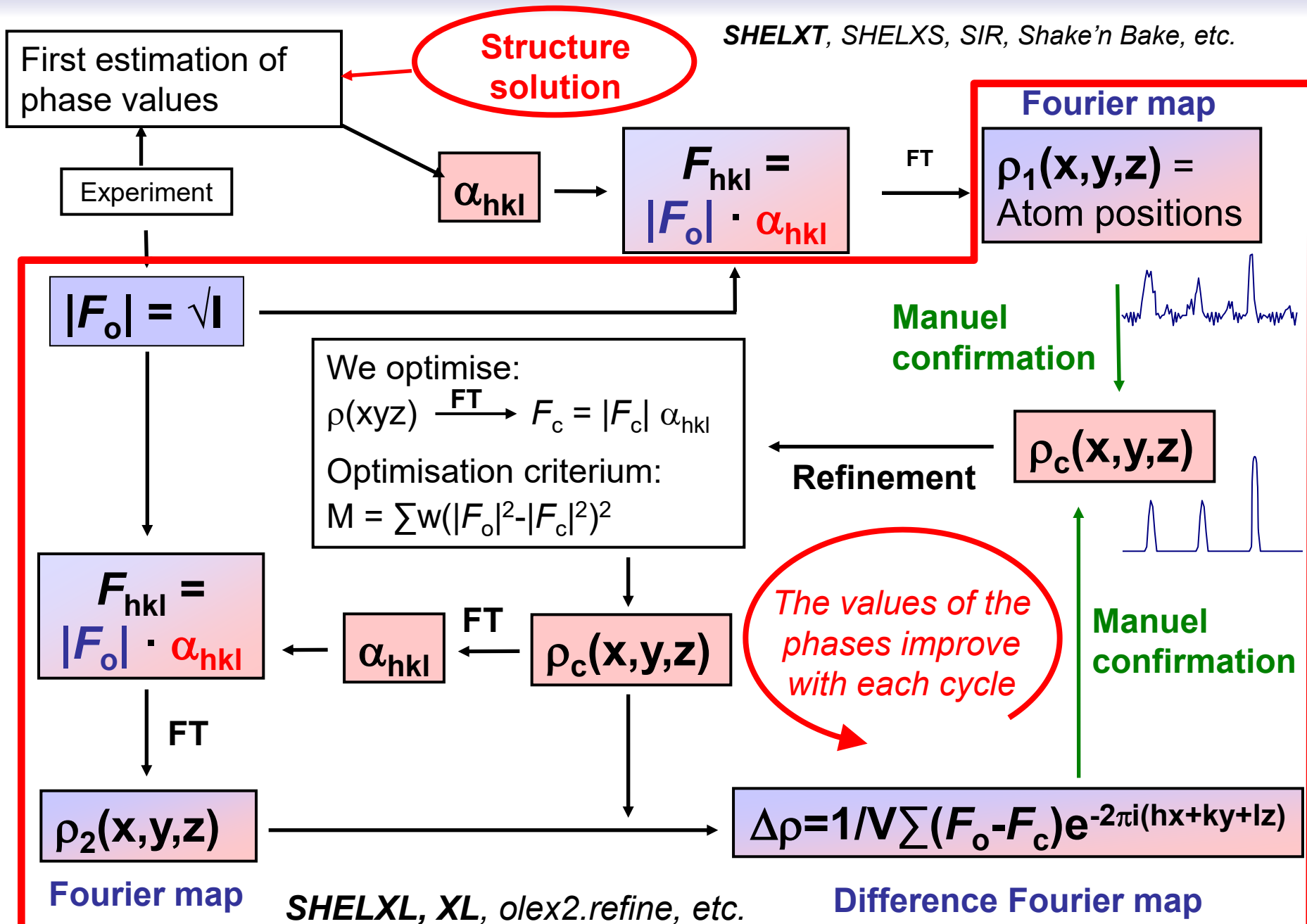


TITL bfs79_a.res in P-1
 CELL 0.71073 9.5892 10.8264 11.3959 111.904 96.814 100.727
 ZERR 2.000 0.0008 0.0010 0.0009 0.006 0.007 0.007
 LATT 1
 SFAC C H B F S
 UNIT 44 16 2 30 2
 TEMP -43
 SIZE 0.1 0.2 0.4
 L.S. 10
 BOND
 LIST 6
 FMAP 2
 PLAN 20
 S001 5 0.72244 0.27584 0.78132 11.00000 0.02851 16.56
 F002 4 0.44727 0.42232 0.86239 11.00000 0.03521 9.16
 F003 4 0.43235 0.19947 0.41550 11.00000 0.03777 8.92
 [...]
 B013 3 0.63372 0.37475 0.67629 11.00000 0.02391 5.24
 HKLF 4
 END

******.RES**

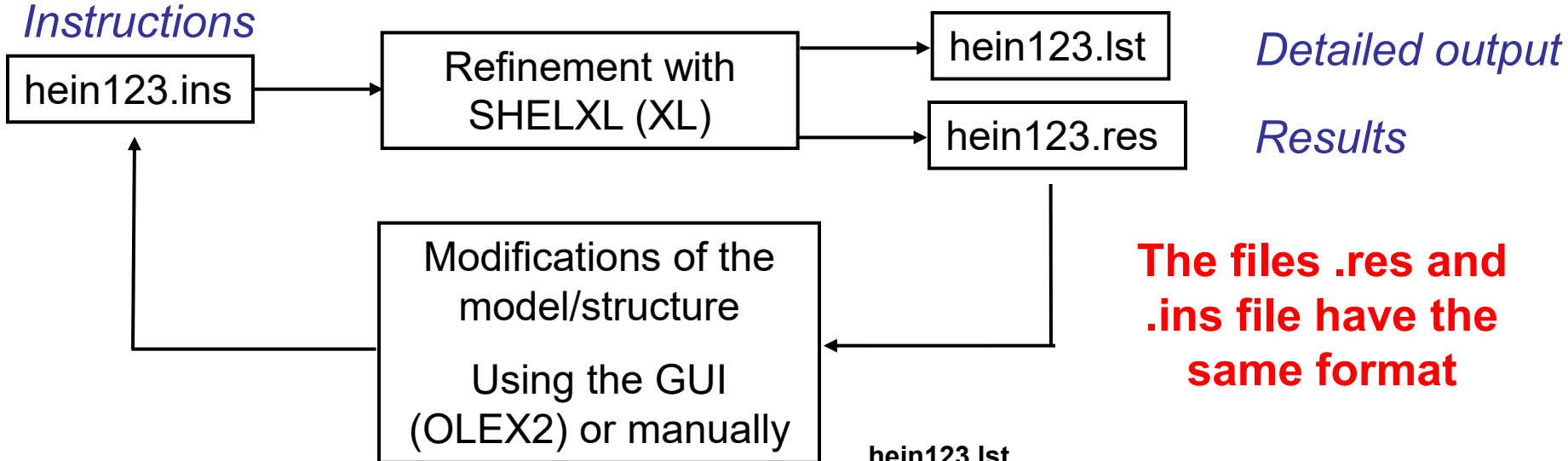
atom positions

Structure refinement



.ins/.res and ***.lst

Instructions



The files .res and .ins file have the same format

hein123.res/hein123.ins

```
TITL Hein123
CELL 0.71073 8.1380 15.4444 15.1323 90.000 98.922 90.000
ZERR 2.00 0.0016 0.0031 0.0030 0.000 0.030 0.000
LATT 1
SYMM -X, 0.5+Y, -Z
SFAC C H N SI ZR
UNIT 70 108 4 8 4
L.S. 6
WGHT 0.100000
FVAR 0.09895
ZR1 5 0.42161 0.25000 0.91998 10.50000 0.02140
SI1 4 0.16560 0.14717 0.71608 11.00000 0.03452
N1 3 0.16921 0.25000 0.66987 10.50000 0.02912
C1 1 0.34492 0.13306 0.80705 11.00000 0.02231
...
HKLF 4
END
```

hein123.lst

50091 Reflections read, of which 26152 rejected

39 Systematic absence violations

0 Inconsistent equivalents

4013 Unique reflections, of which 0 suppressed

R(int) = 0.0363 R(sigma) = 0.0207

Least-squares cycle 1

wR2 = 0.4060 before cycle 1 for 4013 data and 189 /
189 parameters, GooF = S = 7.187; Restrained

GooF = 7.187 for 0 restraints

N	value	esd	shift/esd	parameter
---	-------	-----	-----------	-----------

1	0.36649	0.00352	-210.419	OSF
---	---------	---------	----------	-----

2	0.00205	0.00032	2.885	EXTI
---	---------	---------	-------	------

5	0.02660	0.00068	3.314	U22 Zr1
---	---------	---------	-------	---------

Mean shift/esd = 1.629 Maximum = -210.419 for OSF

Max. shift = 0.029 Å for C25 Max. dU = 0.003 for C14

.ins/.res are constantly overwritten !



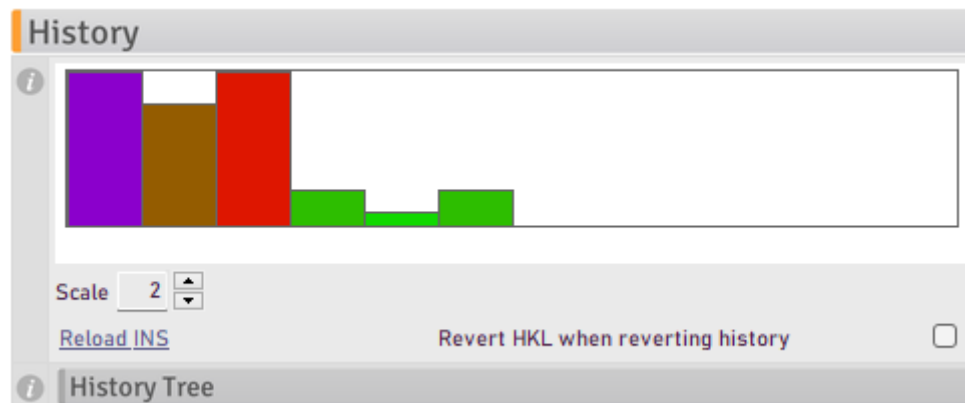
Changes made in the GUI are stored internally by OLEX2.

As soon as you click « Refine » or type « edit ins », OLEX2 updates the ins file with your changes.



olex2	2024-05-30 9:35 AM	File folder	
BFS79.fcf	2024-05-30 10:47 AM	FCF File	130 KB
bfs79.hkl	2024-05-30 9:35 AM	HKL File	226 KB
BFS79.ins	2024-05-30 10:47 AM	INS File	3 KB
BFS79.lst	2024-05-30 10:47 AM	LST File	39 KB
BFS79.lxt	2024-05-30 9:35 AM	LXT File	8 KB
Name	Date modified	Type	Size
originals	2024-05-30 9:28 AM	File folder	
temp	2024-05-30 10:47 AM	File folder	
BFS79.hist5	2024-05-30 10:47 AM	HIST5 File	133 KB
BFS79.metacif	2024-05-30 10:12 AM	METACIF File	1 KB
BFS79.odb	2024-05-30 9:30 AM	ODB File	11 KB
BFS79.phil	2024-05-30 10:12 AM	PHIL File	2 KB

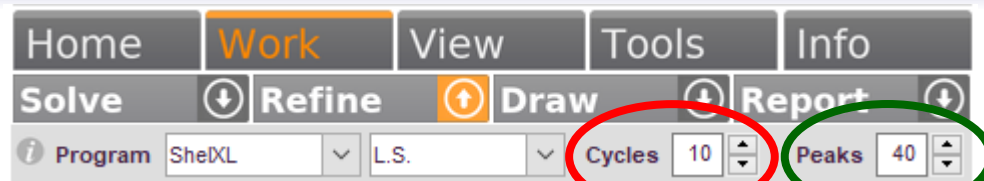
OLEX2 stores previous versions in a hidden directory. You can return to previous versions in WORK-History



Our .res now contains refinement commands

SHELXT

*.res



```
TITL bfs79_a.res in P-1
CELL 0.71073 9.5892 10.8264 11.3959 111.904 96.814 100.727
ZERR 2.000 0.0008 0.0010 0.0009 0.006 0.007 0.007
```

```
LATT 1
SFAC C H B F S
UNIT 44 16 2 30 2
```

```
TEMP -43
SIZE 0.1 0.2 0.4
```

```
L.S. 10
BOND $H
```

```
LIST 6
FMAP 2
```

```
PLAN 40
WGHT 0.100000
```

```
FVAR 0.46325
```

```
S1 5 0.72244 0.27584 0.78132 11.00000 0.02851
```

```
F2 4 0.44727 0.42232 0.86239 11.00000 0.03521
```

```
F3 4 0.43235 0.19947 0.41550 11.00000 0.03777
```

```
[...]
```

```
B1 3 0.63372 0.37475 0.67629 11.00000 0.02391
```

```
HKLF 4
END
```

Elements present
in the unit cell

Include distances
and angles in .lst/.cif

What type of output
(lst, fcf) to produce
(see SHELXL manual)

Number of electron
densities (Q peaks)
in difference Fourier
map

Atom label

Which element?
(number in SFAC)

How many
refinement cycles

What type of Fourier
map to produce. No
need to change this ever.

Weighing scheme

Variables to refine
Variable #1 is the OSF
(overall scale factor)

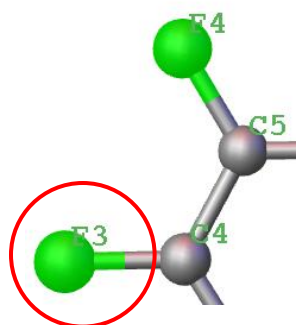
Very important!

S1 1 ... is a carbon atom

C13 5 ... is a sulfur atom

OLEX and the ins file

We can edit the ins file directly from OLEX2 via « edit ins ». When we save & close the ins file, OLEX2 updates the structure. If we do changes in OLEX2, it updates the ins file.



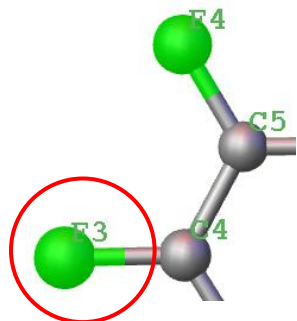
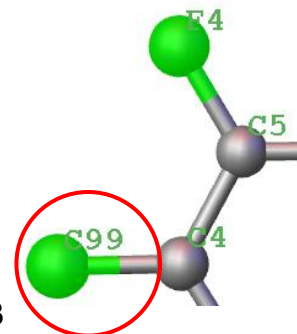
SFAC C H B F S

[...]

F3 4 0.01098 0.15455 0.57775 11.00000 0.05193

Atom type **4** identifies the atom as a fluorine atom.
Changing the label to C99 does not change this.

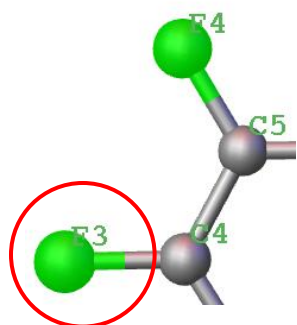
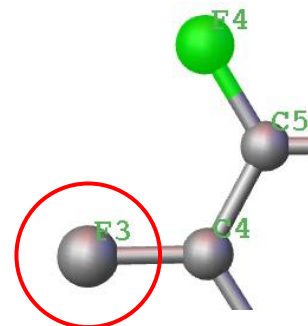
→ **C99** 4 0.01098 0.15455 0.57775 11.00000 0.05193



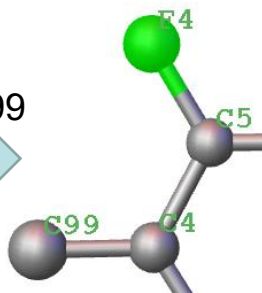
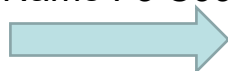
F3 **4** 0.01098 0.15455 0.57775 11.00000 0.05193

Changing the atom type to **1** makes this a carbon atom, even if the label does not change.

→ F3 **1** 0.01098 0.15455 0.57775 11.00000 0.05193

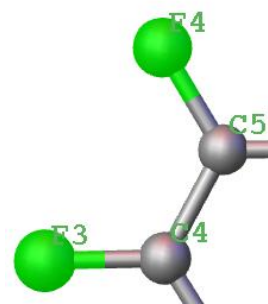


OLEX2:
Name F3 C99



F3 4 [...]

C99 1 [...]



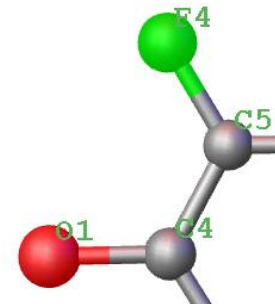
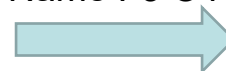
SFAC C H B F S

UNIT 44 16 2 30 2

[...]

F3 **4** 0.01098 [...]

OLEX2:
Name F3 O1



SFAC C H B F S **O**

UNIT 44 16 2 30 2 **1**

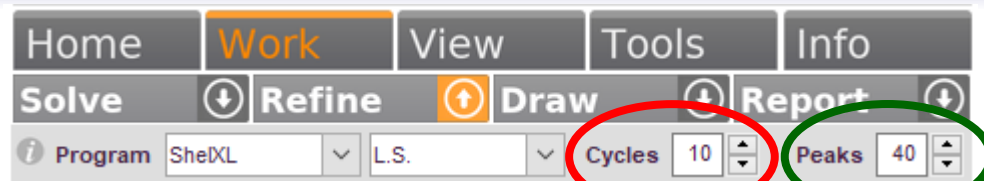
[...]

O1 **6** 0.01098 [...]

Our .res now contains refinement commands

SHELXT

*.res



```
TITL bfs79_a.res in P-1
CELL 0.71073 9.5892 10.8264 11.3959 111.904 96.814 100.727
ZERR 2.000 0.0008 0.0010 0.0009 0.006 0.007 0.007
LATT 1
SFAC C H B F S
```

```
UNIT 44 16 2 30 2
TEMP -43
SIZE 0.1 0.2 0.4
```

```
L.S. 10
BOND $H
```

```
LIST 6
FMAP 2
```

```
PLAN 40
WGHT 0.100000
```

```
FVAR 0.46325
```

```
S1 5 0.72244 0.27584 0.78132 11.00000 0.02851
```

```
F2 4 0.44727 0.42232 0.86239 11.00000 0.03521
F3 4 0.43235 0.19947 0.41550 11.00000 0.03777
[...]
```

```
B1 3 0.63372 0.37475 0.67629 11.00000 0.02391
```

```
HKLF 4
END
```

Elements present
in the unit cell

Include distances
and angles in .lst/.cif

What type of output
(lst, fcf) to produce
(see SHELXL manual)

Number of electron
densities (Q peaks)
in difference Fourier
map

Atom label

How many
refinement cycles

Atom position

Occupation factor

What type of Fourier
map to produce. No
need to change this ever.

Weighing scheme

Variables to refine
Variable #1 is the OSF
(overall scale factor)

U_{iso} value of isotropic
movement

Which element?
(number in SFAC)

Very important!

S1 1 ... is a carbon atom

C13 5 ... is a sulfur atom

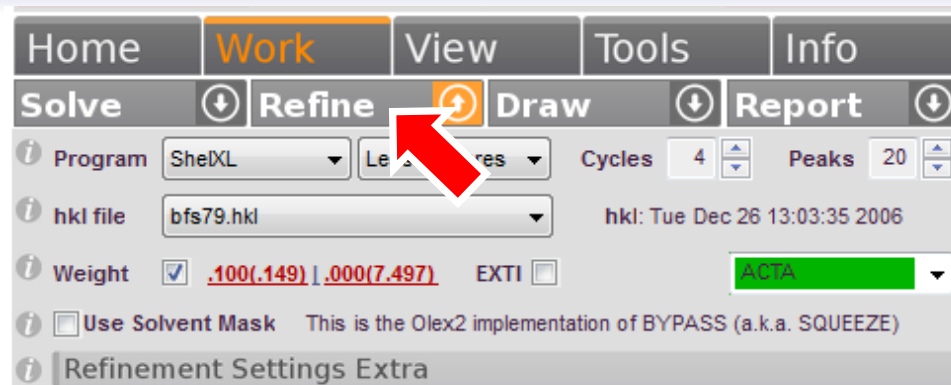
Structure refinement

We optimise:

$$\rho(\text{xyz}) \xrightarrow{\text{FT}} F_c = |F_c| \alpha_{\text{hkl}}$$

Optimization criterium:

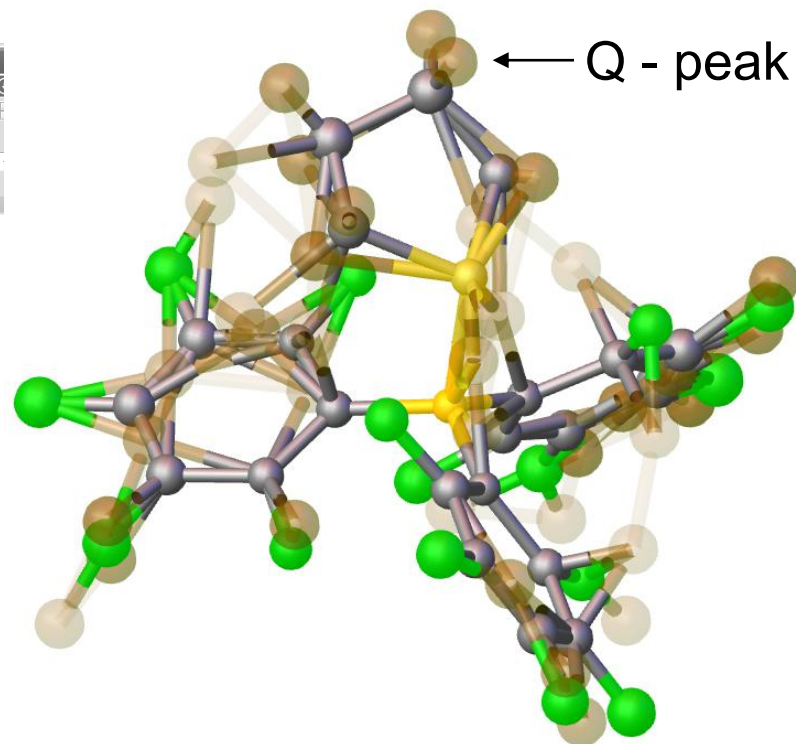
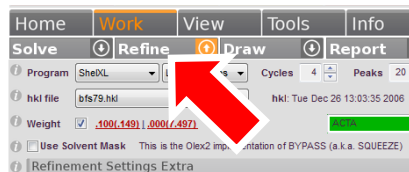
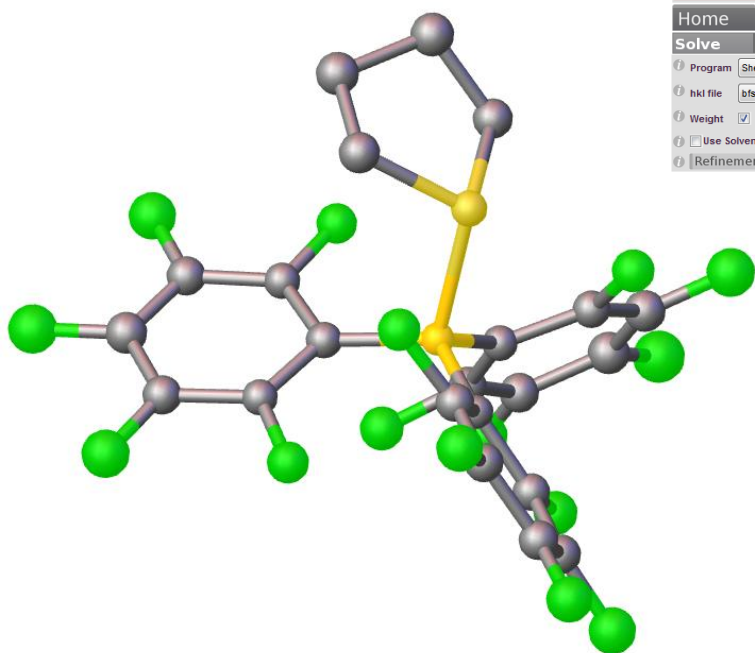
$$M = \sum w(|F_o|^2 - |F_c|^2)^2$$



```

Max. shift = 0.027 A for C9      Max. dU = -0.011 for B1
R2 = 0.4010 before cycle 2 for 4359 data and 157 / 157 parameters
GoF = S = 1.311; Restrained GoF = 1.311 for 0 restraints
Mean shift/esd = 1.662 Maximum = -15.000 for U11 S1 at 16:45:13
Max. shift = 0.027 A for C6      Max. dU = -0.010 for B1
wR2 = 0.3268 before cycle 3 for 4359 data and 157 / 157 parameters
GoF = S = 1.098; Restrained GoF = 1.098 for 0 restraints
Mean shift/esd = 1.585 Maximum = -12.926 for U11 S1 at 16:45:13
Max. shift = 0.022 A for F9      Max. dU = -0.006 for C7
wR2 = 0.3030 before cycle 4 for 4359 data and 157 / 157 parameters
GoF = S = 1.039; Restrained GoF = 1.039 for 0 restraints
Mean shift/esd = 0.217 Maximum = 1.508 for U11 F3 at 16:45:13
Max. shift = 0.006 A for F9      Max. dU = 0.002 for F3
wR2 = 0.3021 before cycle 5 for 4359 data and 0 / 157 parameters
GoF = S = 1.034; Restrained GoF = 1.034 for 0 restraints
wR2 = 0.3021, GoF = S = 1.034, Restrained GoF = 1.034 for all data
R1 = 0.1145 for 3937 Fo > 4sig(Fo) and 0.1221 for all 4359 data
0 atoms may be split and 0 atoms NPD
R1 = 0.1219 for 4359 unique reflections after merging for Fourier
Highest peak 1.48 at 0.2336 0.0348 0.5215 [ 0.61 A from F3 ]
Deepest hole -1.69 at 0.1966 0.1020 0.5312 [ 0.59 A from F3 ]
    
```

Enter « lines 20 » if you want to see more text lines.



TITL bfs79_a.res in P-1
[...]

BFS79.ins

S1	5	0.72250	0.2				
B1	3	0.63358	0.3				
F1	4	0.43229	0.1				
F2	4	0.15239	0.10107	0.37643	11.00000	0.04742	
F3	4	0.01098	0.15455	0.57775	11.00000	0.05193	
F4	4	0.16550	0.31956	0.82027	11.00000	0.04730	
F5	4	0.44702	0.42274	0.86238	11.00000	0.03428	
F6	4	0.77260	0.55856	0.55811	11.00000	0.04160	
F7	4	0.89230	0.48432	0.35458	11.00000	0.04796	
F8	4	0.90576	0.22039	0.23223	11.00000	0.04544	

[...]

What is actually optimized?

Why do we still see Q peaks if all atoms are found?



TITL bfs79_a.res in P-1
[...]

BFS79.res

F2	4	0.152391	0.101074	0.376430	11.00000	0.04742	
F3	4	0.010979	0.154550	0.577751	11.00000	0.05193	

[...]
END

Q1	1	0.7707	0.9637	0.9808	11.00000	0.05	1.65
Q2	1	0.7235	0.8144	0.8991	11.00000	0.05	1.10
Q3	1	0.6140	0.8724	0.8755	11.00000	0.05	1.10

[...]

Which parameters are refined ?

File .ins:

CELL 0.71073 8.1380 15.4444 15.1323 90 **98.922** 90

Si1 4 **0.16560 0.14717 0.71608** 11.00000 0.03452

Atomic position

The atomic positions are provided as **fractional coordinates relative to the crystal system**, i. e. the unit cell dimensions. Thus, the values **are not in Å**. For example, the atom Si1 is found at :

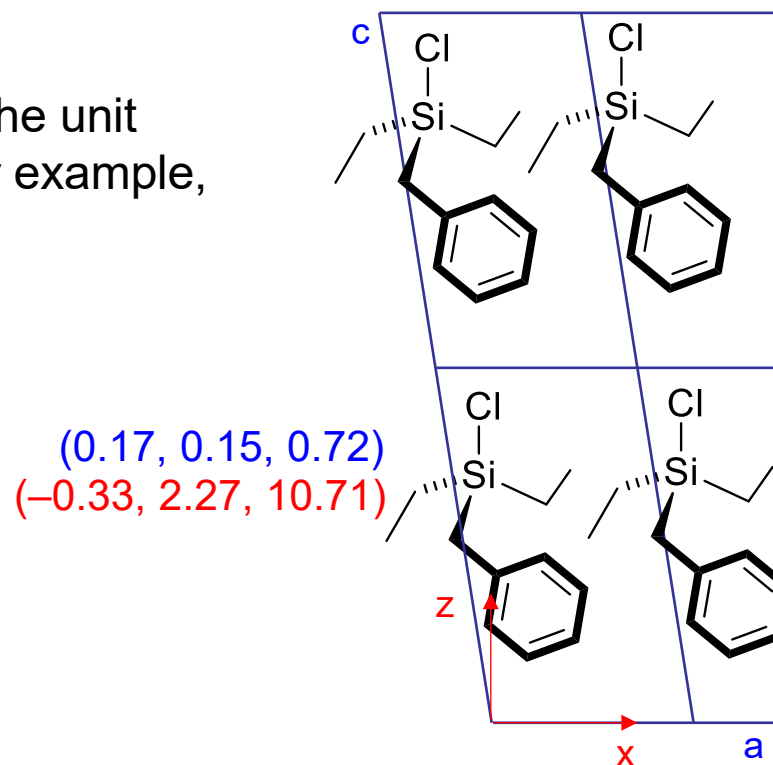
$$x = u \cdot a + v \cdot c \cdot \cos\beta = -0.333 \text{ Å}$$

$$y = v \cdot b = 2.273 \text{ Å}$$

$$z = w \cdot c \cdot \sin\beta = 10.705 \text{ Å}$$

The symbols for the **fractional atomic positions** are typically the letters **u**, **v**, and **w**.

Given that each atoms is inside the unit cell, *u*, *v*, and *w* have values between 0 and 1. For practical reasons (connected set of coordinates), we find sometimes values <0 or >1, **but there is no justification for values <-1 or >2**.



Which parameters are refined ?

SI1	4	0.16560	0.14717	0.71608	1	1.00000	0.03452
-----	---	---------	---------	---------	---	----------------	---------

Occupation
factor

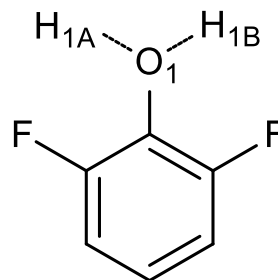
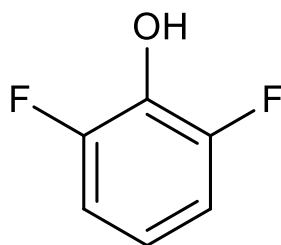
The occupation factor indicates how many atoms occupy this position (or better: which percentage of these positions is occupied by an atom). Typically, this value is 1. It is not possible to have more than one atom per position, but it is possible to have smaller values.

Disorder :

*The 1 indicates that the occupation factor is not free to refine (= fixed).
Freely refining the occupation factor is (normally) nonsense.*

***.ins:

O1	3	0.12560	0.23453	0.83456	1	1.00000	0.02932
H1A	2	0.12864	0.23364	0.80236	1	0.50000	0.04732
H2A	2	0.12853	0.23923	0.85935	1	0.50000	0.04593



*You can freeze any
variable by adding
a 1 in front.*

Thermal movement / The temperature factor

Atomic position				Occupation factor	U_{iso}	
SI1	4	0.16560	0.14717	0.71608	11.00000	0.03452

U_{iso} ?

- The spatial distribution of the electron density is influenced by the thermal motion of the atoms.
- Thermal motion is not identical (in amplitude and direction) for all atoms.
- The timescale of a diffraction experiment is much longer than thermal motion.

⇒ **We obtain a distribution of electron density,
averaged over time and all unit cells**

Isotropic motion:

- The motion (vibration) of an atom is identical in all directions.
- It can be described by a Gaussian distribution:

$$\rho(r') = \sqrt{2\pi U} e^{\frac{-r'^2}{2U}}$$

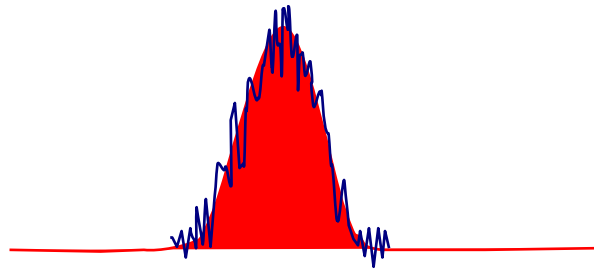
ρ : electron density

r' : distance of the atom from its equilibrium position

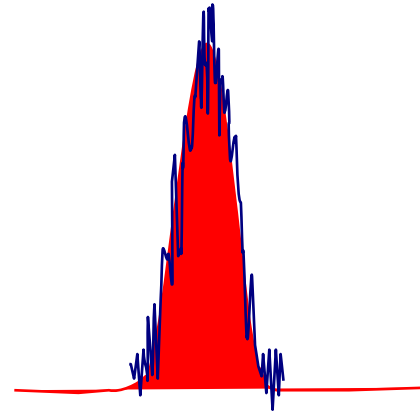
$U = \langle r'^2 \rangle$ Average of the squared displacement [$\text{\AA}^2$]

$B = 8\pi^2 U$ Atomic temperature factor (Debye-Waller factor) [$\text{\AA}^2$]

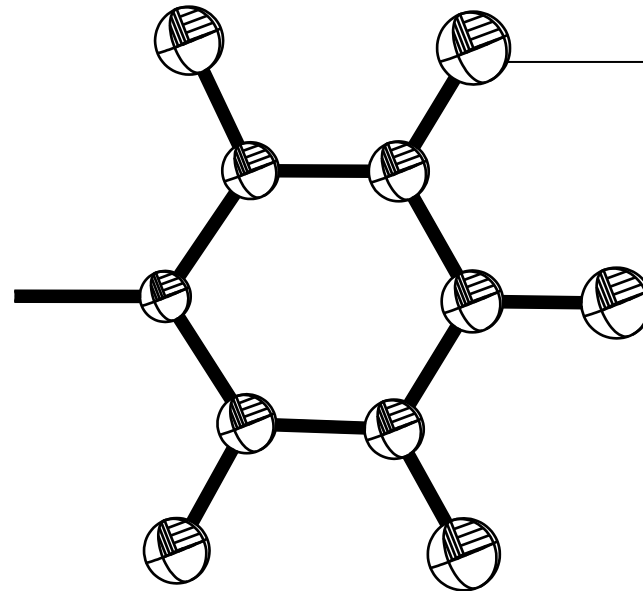
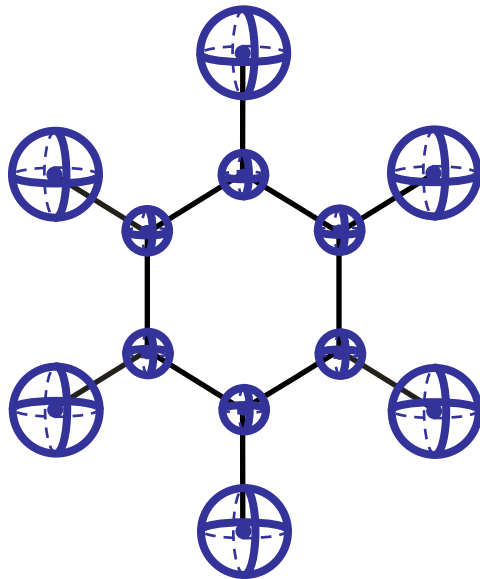
Thermal movement / The temperature factor



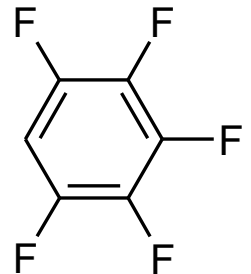
$$U_{\text{iso}} = 0.05 \text{ \AA}^2$$



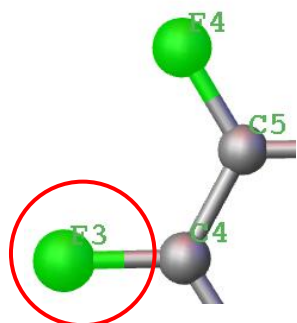
$$U_{\text{iso}} = 0.03 \text{ \AA}^2$$



Probability of x%
(typically 50%)
that the atom is
found inside a
sphere with the
radius indicated
here.



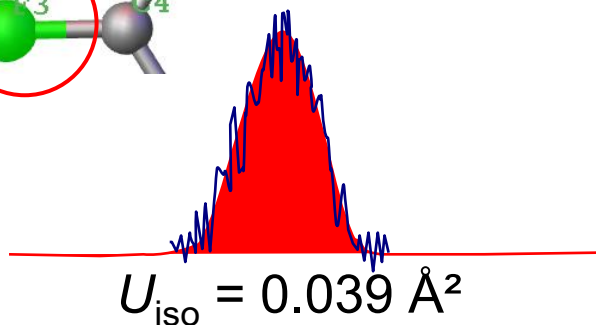
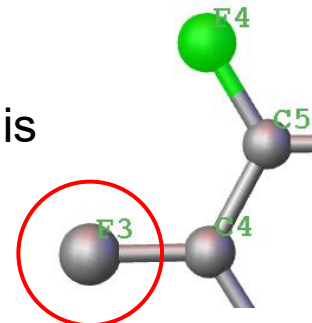
The wrong atom type and U_{iso}



F3 **4** 0.01098 0.15455 0.57775 11.00000 0.05193

Let's change the atom to a carbon atom. Attention, it is the second parameter which matters, not the label.

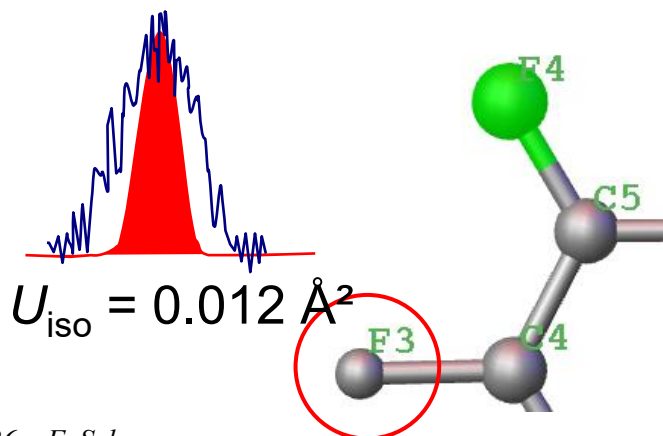
C99 **1** 0.01098 0.15455 0.57775 11.00000 **0.03873**



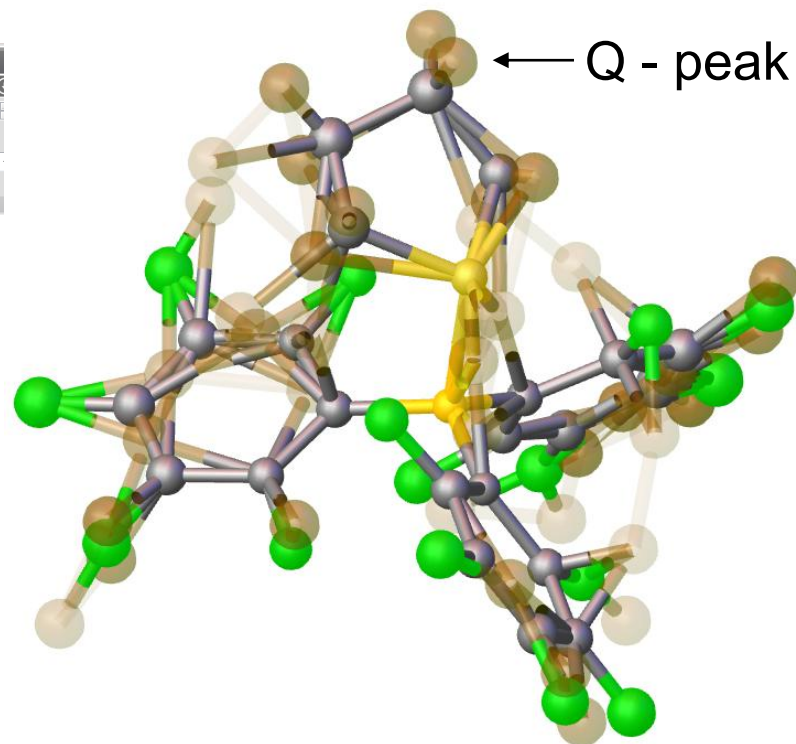
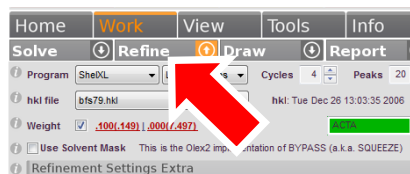
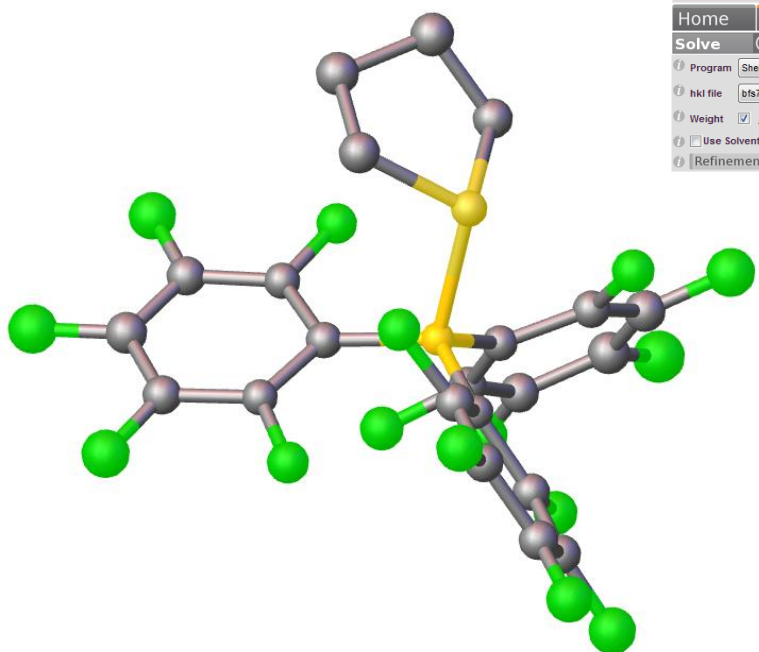
Refinement:



The isotropic parameter shrinks, since our model now has less electrons at this position.



C99 1 0.011198 0.154210 0.577322 11.00000 **0.01259**



TITL bfs79_a.res in P-1

[...]

S1	5	0.72250	0.27589	0.78126	11.00000	0.02810
B1	3	0.63358	0.3			
F1	4	0.43229	0.1			
F2	4	0.15239	0.10107	0.37643	11.00000	0.04742
F3	4	0.01098	0.15455	0.57775	11.00000	0.05193
F4	4	0.16550	0.31956	0.82027	11.00000	0.04730
F5	4	0.44702	0.42274	0.86238	11.00000	0.03428
F6	4	0.77260	0.55856	0.55811	11.00000	0.04160
F7	4	0.89230	0.48432	0.35458	11.00000	0.04796
F8	4	0.90576	0.22039	0.23223	11.00000	0.04544

[...]

BFS79.ins

Why do we still see Q peaks if all atoms are found?



TITL bfs79_a.res in P-1

[...]

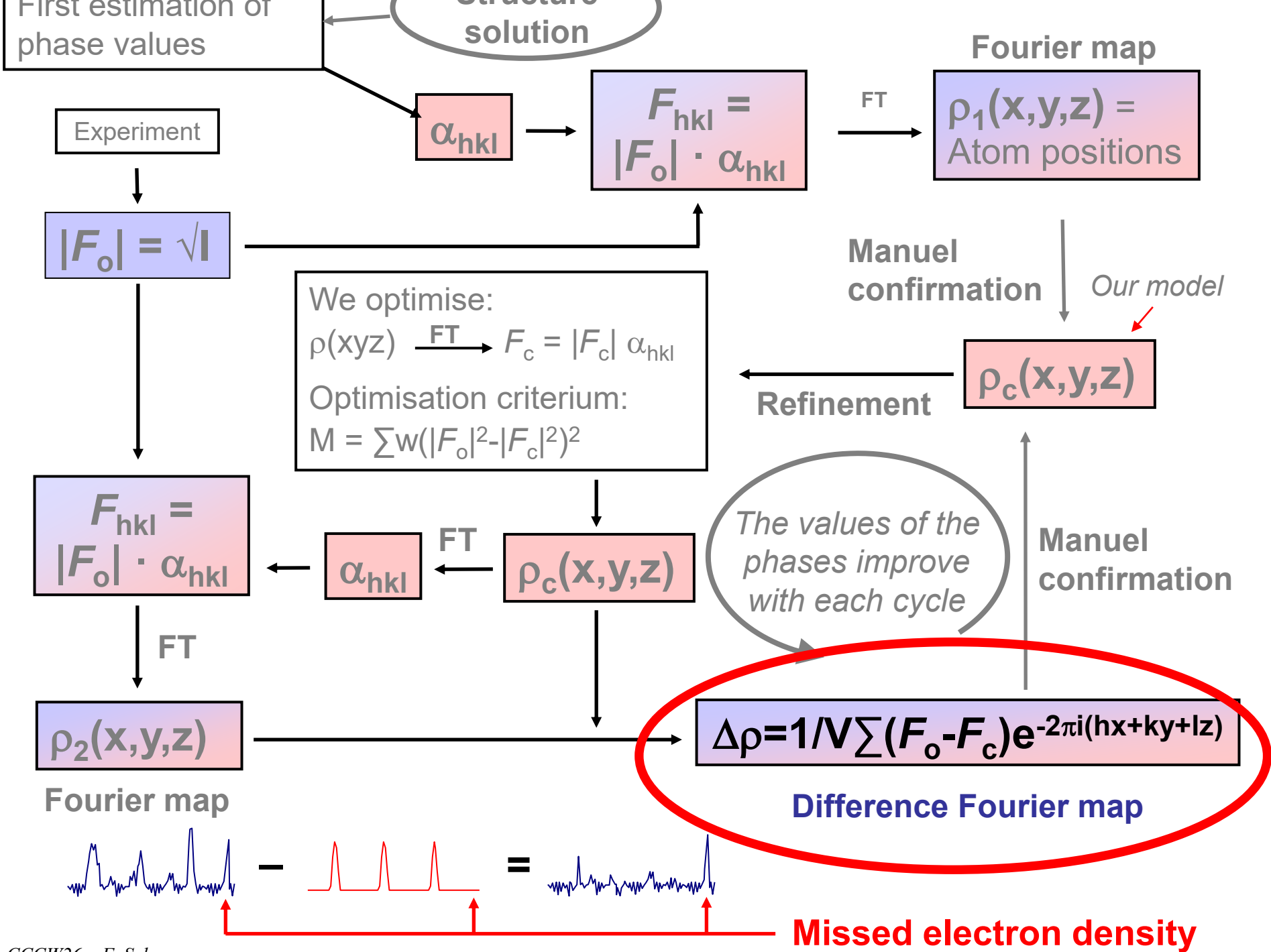
S1	5	0.722498	0.275894	0.781256	11.00000	0.02810
					11.00000	0.02444
					11.00000	0.03683
F2	4	0.152391	0.101074	0.376430	11.00000	0.04742
F3	4	0.010979	0.154550	0.577751	11.00000	0.05193

[...]
END

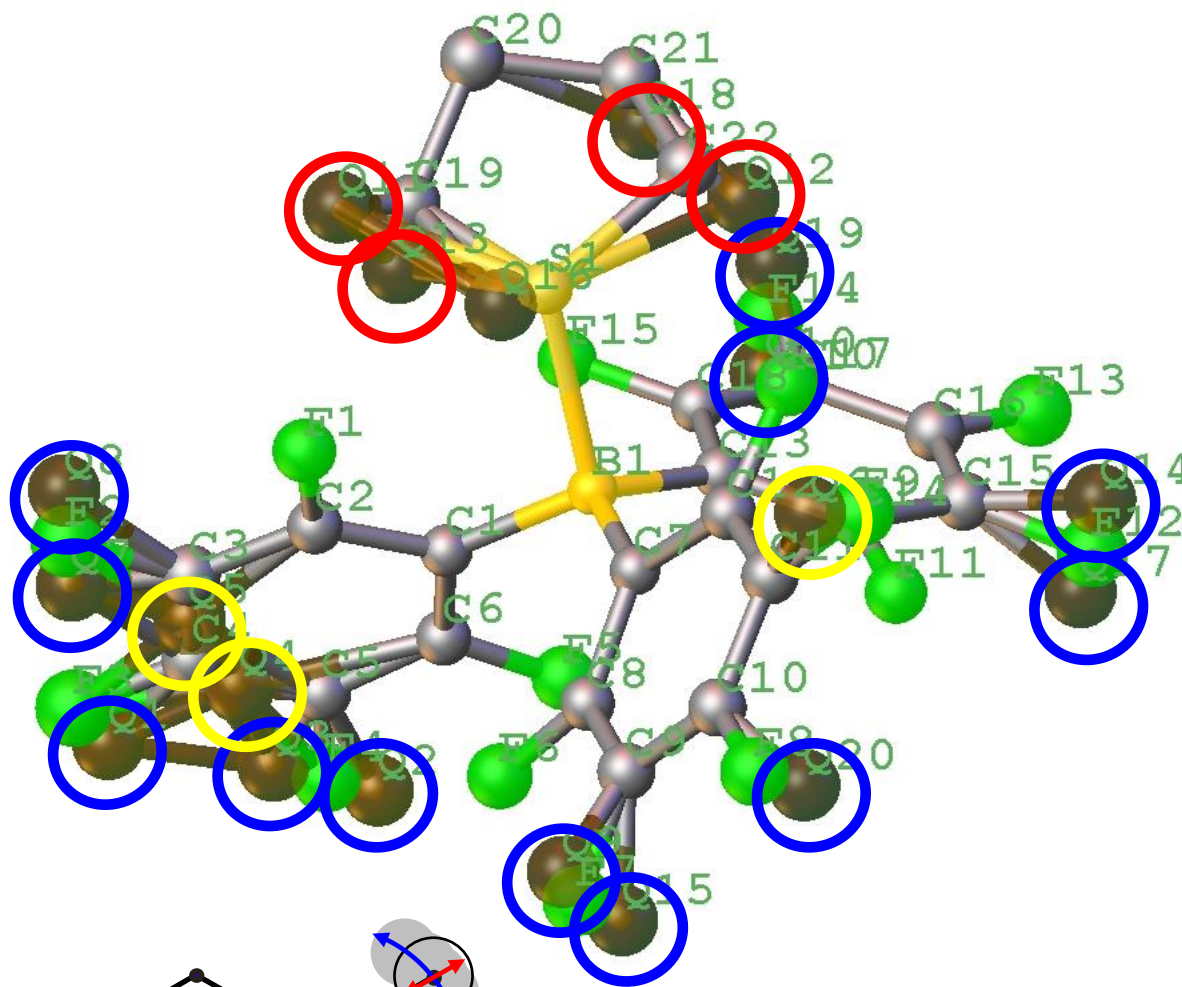
BFS79.res

Q1	1	0.7707	0.9637	0.9808	11.00000	0.05	1.65
Q2	1	0.7235	0.8144	0.8991	11.00000	0.05	1.10
Q3	1	0.6140	0.8724	0.8755	11.00000	0.05	1.10

[...]



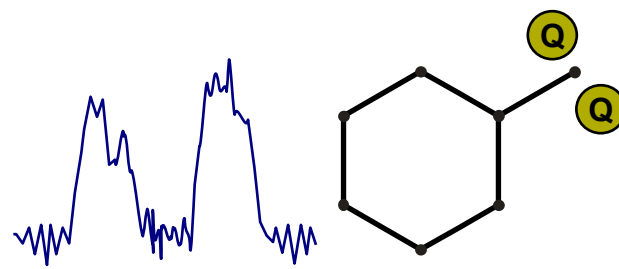
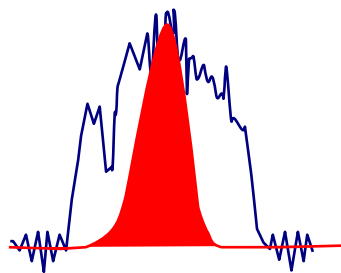
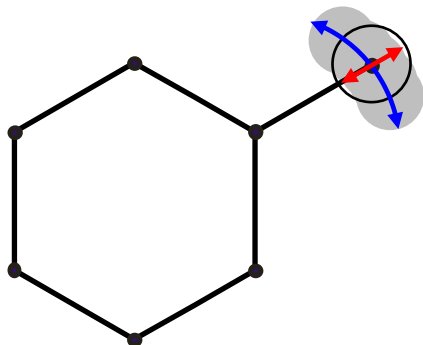
Remaining maxima in the difference Fourier map



All atoms of the structure have been found and remaining maxima are not any longer associated with additional atoms.

The remaining maxima of electron density are due to (in decreasing order of intensity) :

- Thermal movement of atoms
- **Hydrogen atoms**
- **Noise and artefacts**

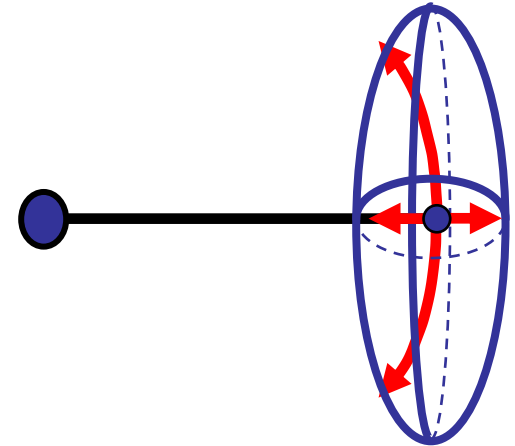


Thermal movement / The temperature factor

Anisotropic thermal motion :

- In reality, thermal movement of atoms is not isotropic. Deformation of bond angles are typically less energetic than stretching vibration parallel to the bond.

Compare: C-H stretch: 3000 cm^{-1} , bending: $600\text{-}1500\text{ cm}^{-1}$.



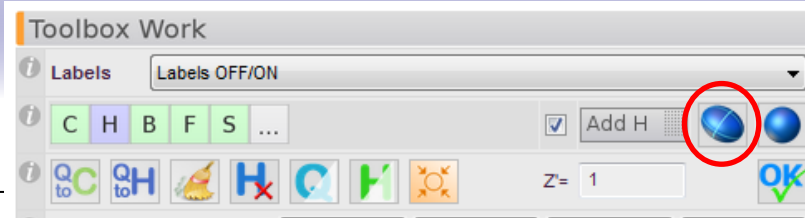
- In our structural model, thermal movement is described by an ellipsoid (three Gaussian functions in three dimensions). The mathematic form is that of a symmetric tensor:

$$U = \begin{pmatrix} \langle x'^2 \rangle & \langle x' y' \rangle & \langle x' z' \rangle \\ \langle x' y' \rangle & \langle y'^2 \rangle & \langle y' z' \rangle \\ \langle x' z' \rangle & \langle y' z' \rangle & \langle z'^2 \rangle \end{pmatrix}$$

$$q(r^*) = e^{-2\pi^2 (U_{11}x'^2 + U_{22}y'^2 + U_{33}z'^2 + 2U_{12}x'y' + 2U_{13}x'z' + 2U_{23}y'z')}$$

- Instead of **1 parameter** (U_{iso}), we now work with **6 parameters** (three values for the three diameters of the ellipsoid, three for its orientation in space).
- Anisotropic refinement thus becomes challenging if the ratio of data/parameters is low.

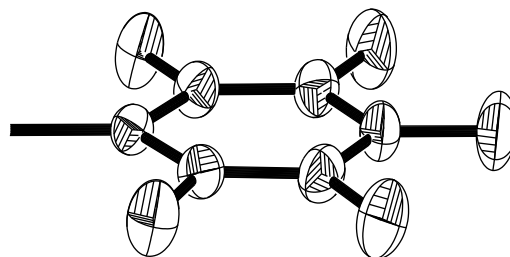
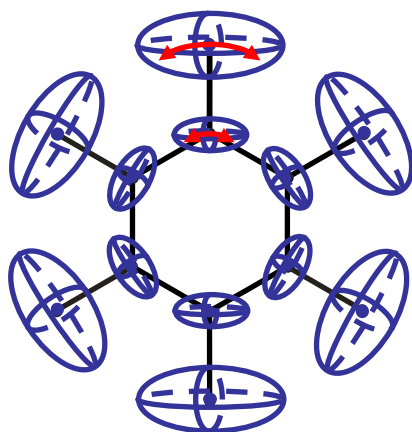
Anisotropic refinement



OLEX2

```
*.ins:
ANIS
WGHT      0.100000
FVAR      0.71880
S1      5      0.277586      0.724064      0.718741      11.00000      0.02858
B1      3      0.366643      0.625301      0.823380      11.00000      0.02507
F1      4      0.079477      0.474487      0.653317      11.00000      0.03965
```

```
*.res:
WGHT      0.100000
FVAR      0.73527
S1      5      0.277751      0.724339      0.718759      11.00000      0.03304      0.02854 =
              0.02460      0.01056      0.00663      0.01163
B1      3      0.366302      0.625285      0.823440      11.00000      0.02772      0.02171 =
              0.02300      0.00834      0.00485      0.00625
F1      4      0.079086      0.473686      0.653072      11.00000      0.02849      0.03982 =
              0.04679      0.01366      0.00319      0.00667
```



U_{eq} is the equivalent isotropic value, calculated from :

$$U_{eq} = \frac{1}{3} \sum_{i=1}^3 \sum_{j=1}^3 U_{ij} a_i^* a_j^* \vec{a}_i \cdot \vec{a}_j = \sqrt[3]{U_1 U_2 U_3}$$

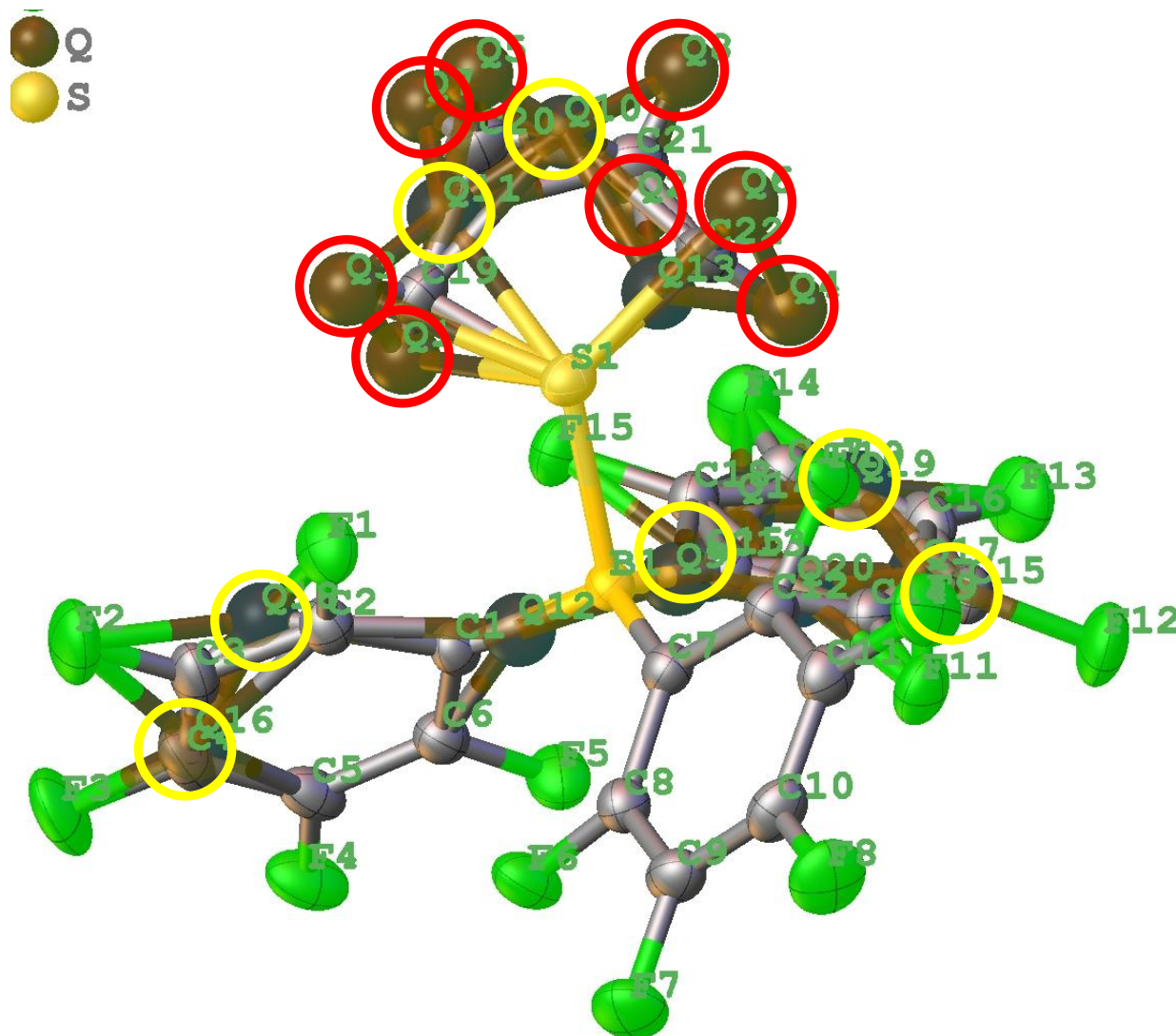
Typical values for U_{eq} :

Heavy atoms: 0.005-0.02 Å²

Light atoms (H-F): 0.01-0.06 Å²

Terminal atoms: 0.03-0.2 Å²

Difference Fourier map after anisotropic refinement

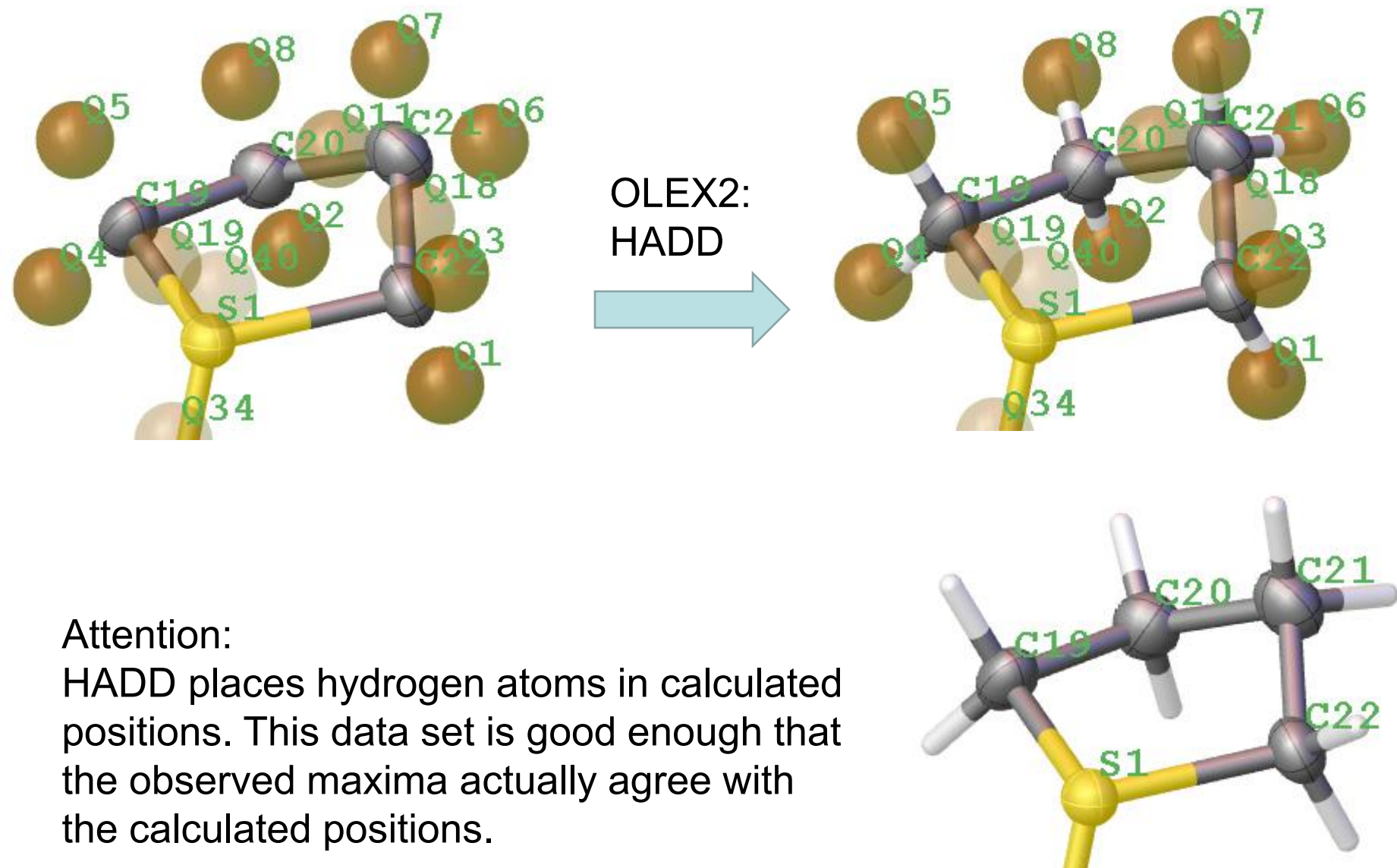


After anisotropic refinement, the remaining maxima can be attributed to:

- Thermal movement of atoms
- Hydrogen atoms
- Noise and artefacts

The 8 most intense maxima (Q peaks with the lowest numbers, i. e. Q1-Q8) are found where we would expect hydrogen atoms!

Adding hydrogen atoms



Attention:
HADD places hydrogen atoms in calculated positions. This data set is good enough that the observed maxima actually agree with the calculated positions.

What happens in the ins file ?

```
C19  1  0.58371 0.12711 0.76751 11.00000 0.05447 0.02329 0.03258 =
0.01158 0.01211 0.00636
C20  1  0.53585 0.16502 0.89601 11.00000 0.04641 0.03599 0.03512 =
0.01616 0.01325 0.01003
```



OLEX2:
HADD

```
C19  1  0.58371 0.12711 0.76751 11.00000 0.05447 0.02329 0.03258 =
0.01158 0.01211 0.00636
```

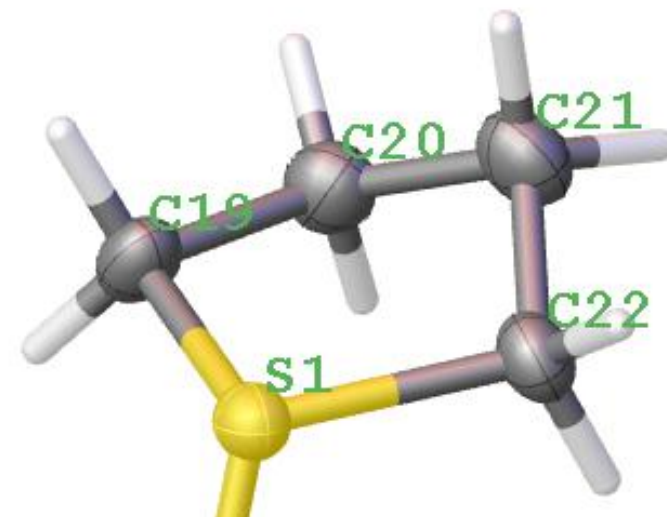
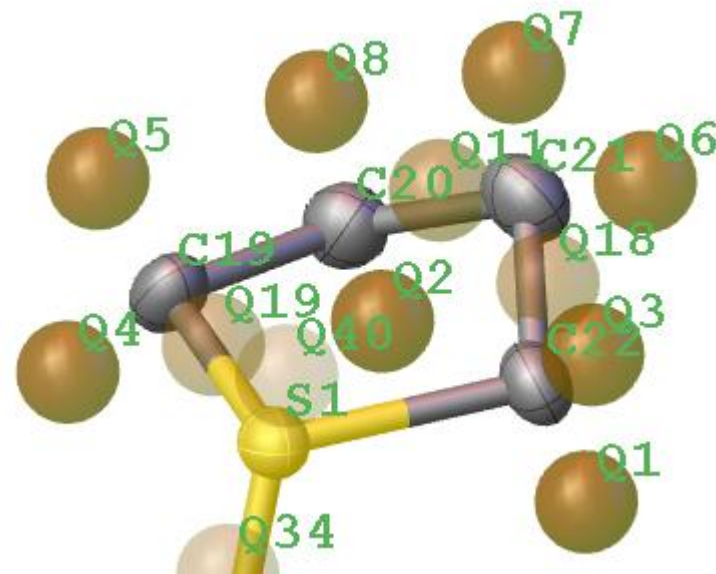
AFIX 23 ← AFIX indicates the start of a « rigid group ». The internal geometry of a rigid group does not change.

```
H19a  2  0.50108 0.10640 0.69784 11.00000 -1.20000
```

```
H19b  2  0.62427 0.04684 0.74958 11.00000 -1.20000
AFIX 0
```

```
C20  1  0.53585 0.16502 0.89601 11.00000 0.04641 0.03599 0.03512 =
0.01616 0.01325 0.01003
```

```
AFIX 23
H20a  2  0.45720 0.21130 0.89603 11.00000 -1.20000
H20b  2  0.50235 0.08214 0.91123 11.00000 -1.20000
AFIX 0
```



Rigid groups in SHELXL

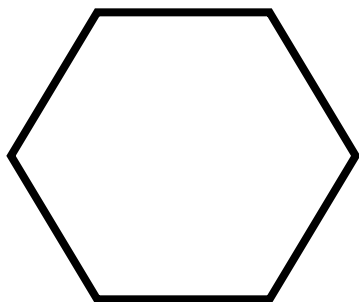
For non-H atoms:

In rigid groups, the atom positions ($3n$) are replaced by the position and orientation (6 variables) of a group of atoms in **an idealized geometry**. The geometry of this group is **invariable** and the atoms cannot move independent during the refinement.

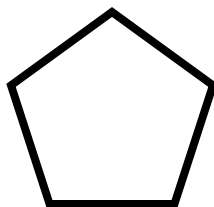
type of rigid group
AFIX *mn* *refinement of the rigid group*

AFIX *m6* : the group is completely rigid

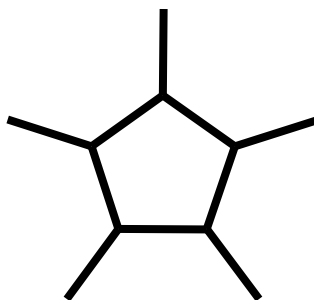
AFIX *m9* : the group can expand or contract, but keeps its relative geometry.



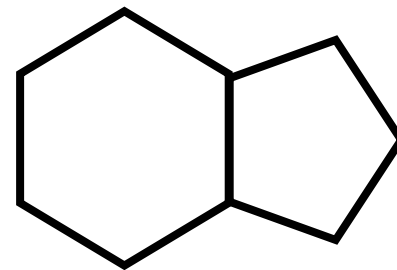
AFIX 66
C1 x y z
:
C6 x y z
AFIX 0



AFIX 56
C1 x y z
:
C5 x y z
AFIX 0



AFIX 106
C1 x y z
:
C10 x y z
AFIX 0

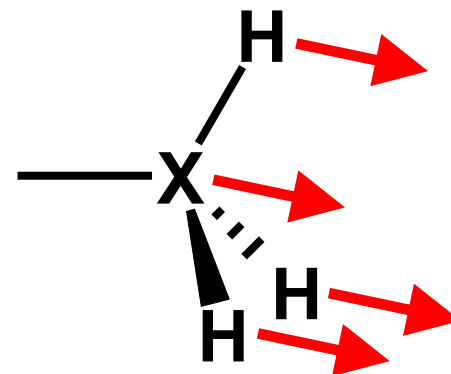


AFIX 116
C1 x y z
:
C11 x y z
AFIX 0

Riding model for hydrogen atoms

The hydrogen atoms “ride” on the atom to which they are bound with a fixed, idealized geometry. During the refinement, the same displacement are applied to the whole group.

$$\begin{aligned}x_H &= x_X + \Delta x \\y_H &= y_X + \Delta y \\z_H &= z_X + \Delta z \\U_{\text{iso},H} &= 1.2/1.5 \cdot U_{\text{eq},X}\end{aligned}$$



If hydrogen atoms are refined using the “riding model”, **no additional parameter** is added to the refinement.

Methyl group

- with freely refined hydrogens : 9 (C) + 3·4 (H isotropic) = 21 parameter
- riding model : 9 parameters

Hydrogen atoms: 4. Constraints / “riding model”

AFIX mn: m = type of the rigid group; n = refinement mode (for more details : see the SHELXL manual)

m = 1 : Y-CH(-Y)₂

2 : Y-CH₂-Y

3 : Y-CH₃

4 : C(sp²/arom.)-H

8 : X-OH

9 : X=CH₂

12 : Y-CH₃ (désordonné)

13 : Y-CH₃ (angle dièdre optimisé)

14 : X-OH (angle dièdre optimisé)

15 : X_{4/5}BH

16 : X≡CH

n = 3 : No refinement.

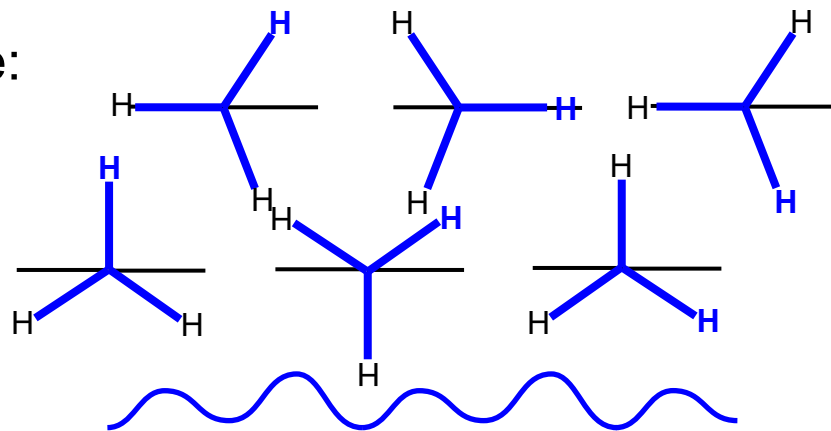
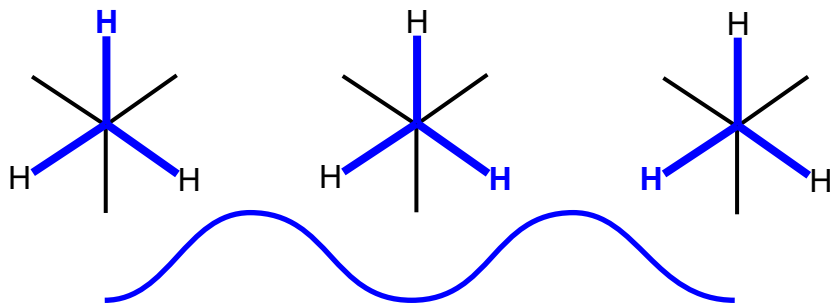
4 : As n=3, but the C-H distance is refined (same for all C-H of that group.)

7 : Only for CH₃: same as n=3, but rotation around the Y-CH₃ is allowed.

8 : same as n=7 with refinement of the C-H distance.

We typically use: AFIX 33, AFIX 23, AFIX 13, AFIX 43, AFIX 137 (or 37), AFIX 127

AFIX 37, 133 or 137 important for C_{sp2}-Me:



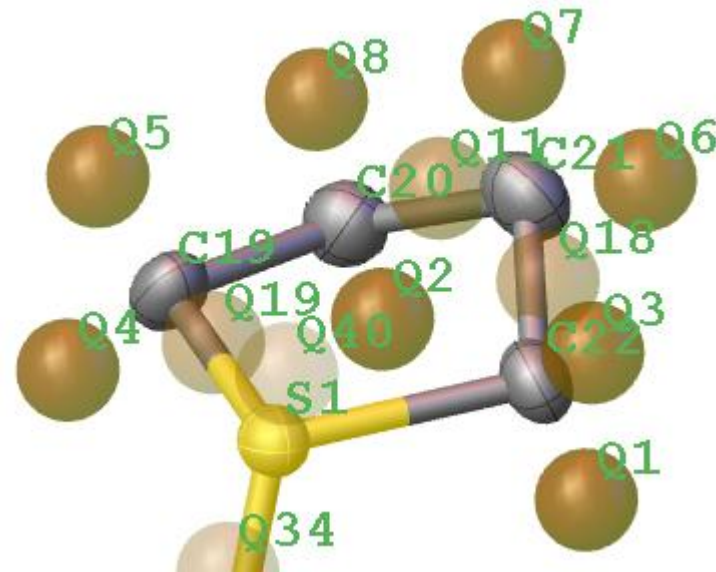
What happens in the ins file ?

```
C19  1  0.58371 0.12711 0.76751 11.00000 0.05447 0.02329 0.03258 =
0.01158 0.01211 0.00636
C20  1  0.53585 0.16502 0.89601 11.00000 0.04641 0.03599 0.03512 =
0.01616 0.01325 0.01003
```

*HADD is very convenient,
but **always** verify what it
did! In particular CH₂ vs.
C_{Ar}H errors*



OLEX2:
HADD



```
C19  1  0.58371 0.12711 0.76751 11.00000 0.05447 0.02329 0.03258 =
0.01158 0.01211 0.00636
```

AFIX 23

↑ rigid group starts

copied
from C19

Calculated :
 $1.2 \cdot U_{eq}(C19)$

```
H19a  2  0.50108 0.10640 0.69784 11.00000 -1.20000
```

calculated from C19 (and neighbours)

```
H19b  2  0.62427 0.04684 0.74958 11.00000 -1.20000
```

AFIX 0

↑ rigid group ends

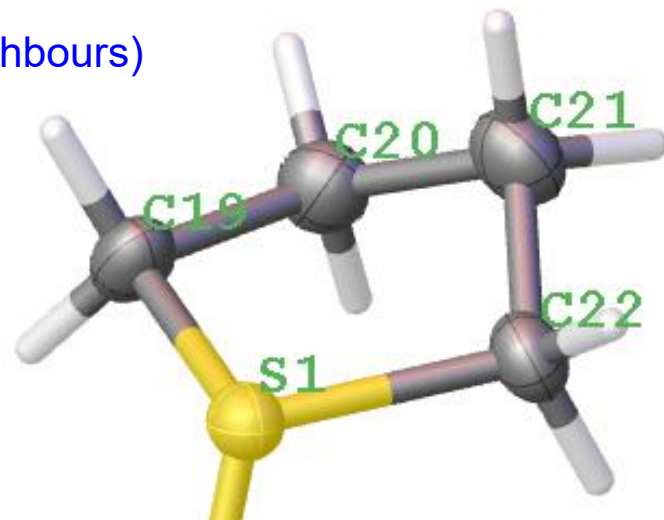
```
C20  1  0.53585 0.16502 0.89601 11.00000 0.04641 0.03599 0.03512 =
0.01616 0.01325 0.01003
```

AFIX 23

```
H20a  2  0.45720 0.21130 0.89603 11.00000 -1.20000
```

```
H20b  2  0.50235 0.08214 0.91123 11.00000 -1.20000
```

AFIX 0



And what's left after that...

After refinement of hydrogen atoms, all electron density maxima are smaller than $0.5 \text{ e}^-/\text{\AA}^3$ and can be ignored. ($0.5 \text{ e}^-/\text{\AA}^3$ is considered the “allowable noise level.”)

INFO Q1 :

Atom	Type	X	Y	Z	Ueq	Peak
Q1	Q	0.458	0.660	0.845	0.050	0.280

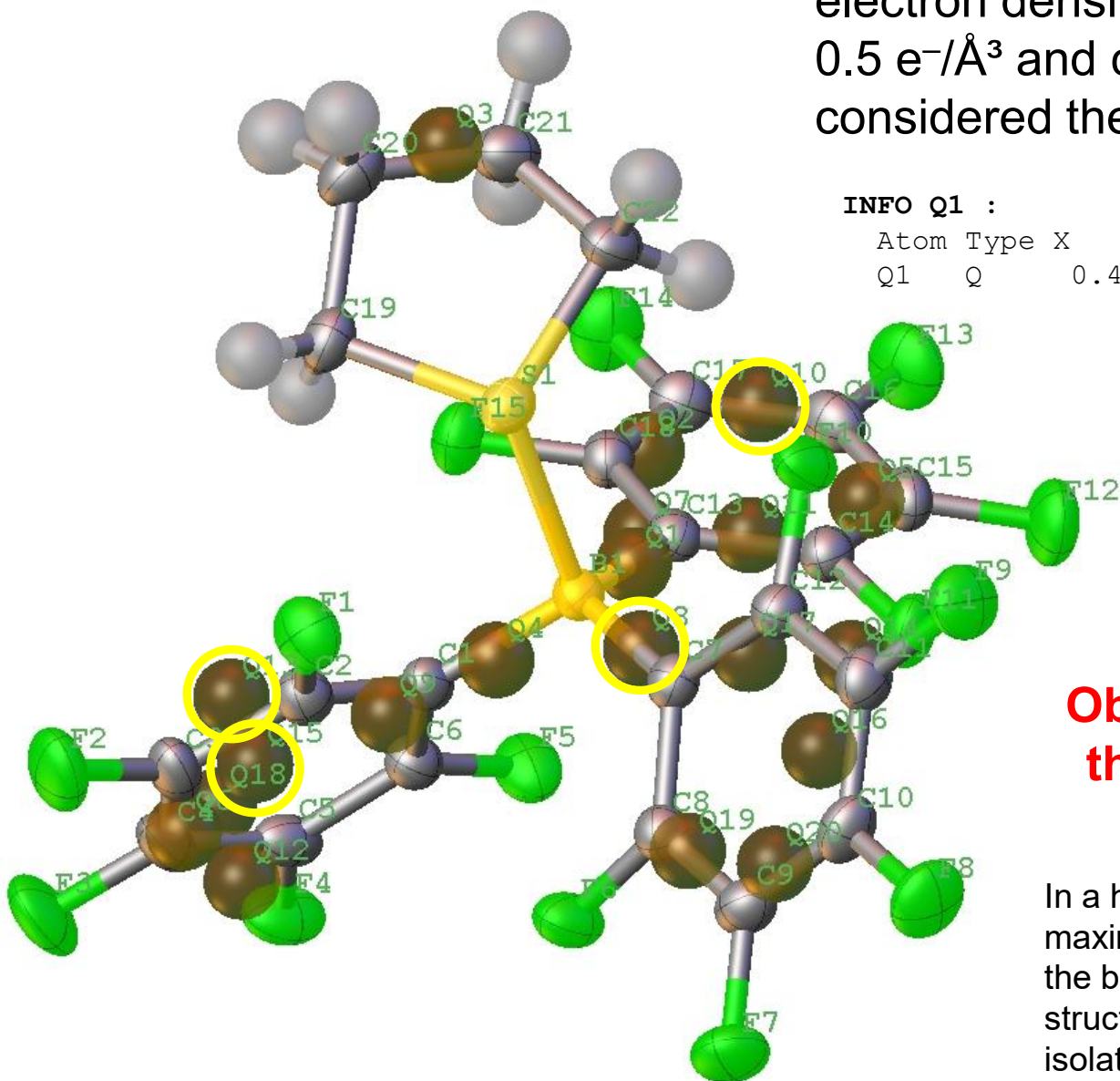
• ~~Thermal movement of atoms~~

• ~~Hydrogen atoms~~

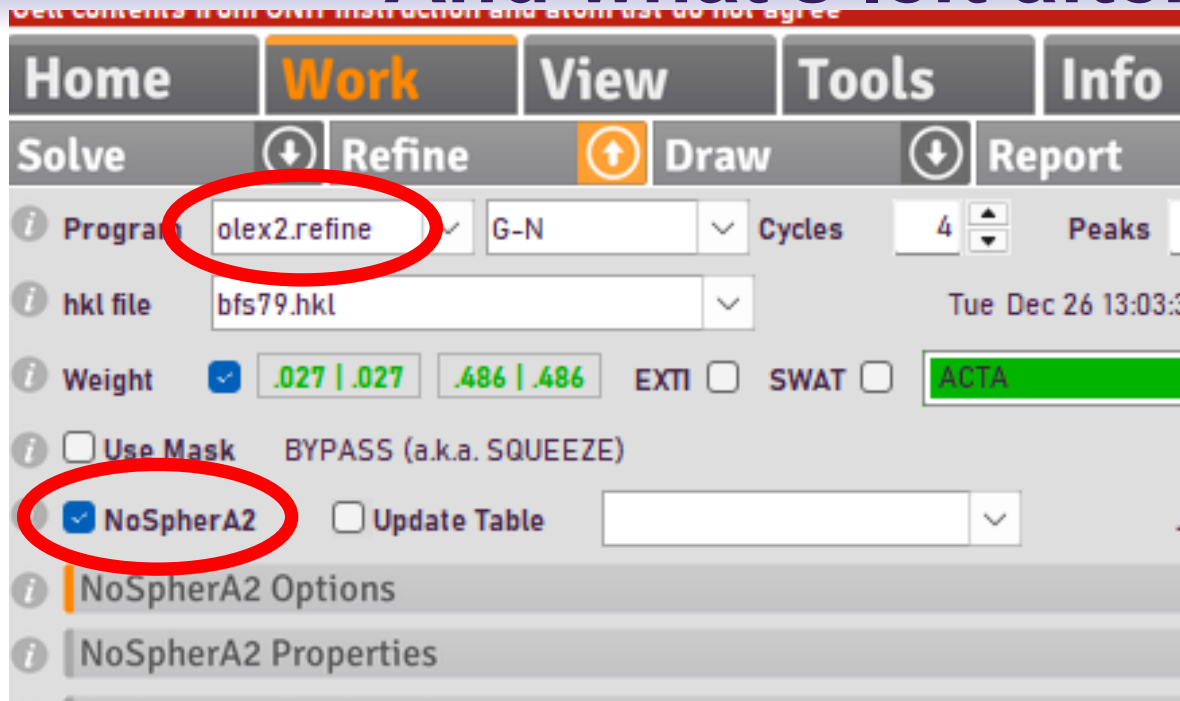
• Noise and artefacts

Observe the positions of the maxima, though! Is this really noise?

In a high-quality dataset, the remaining maxima correspond to electron density in the bonds. This is an artefact of our structural model: we assume a model of isolated, non-interacting, spherical atoms.



And what's left after that...



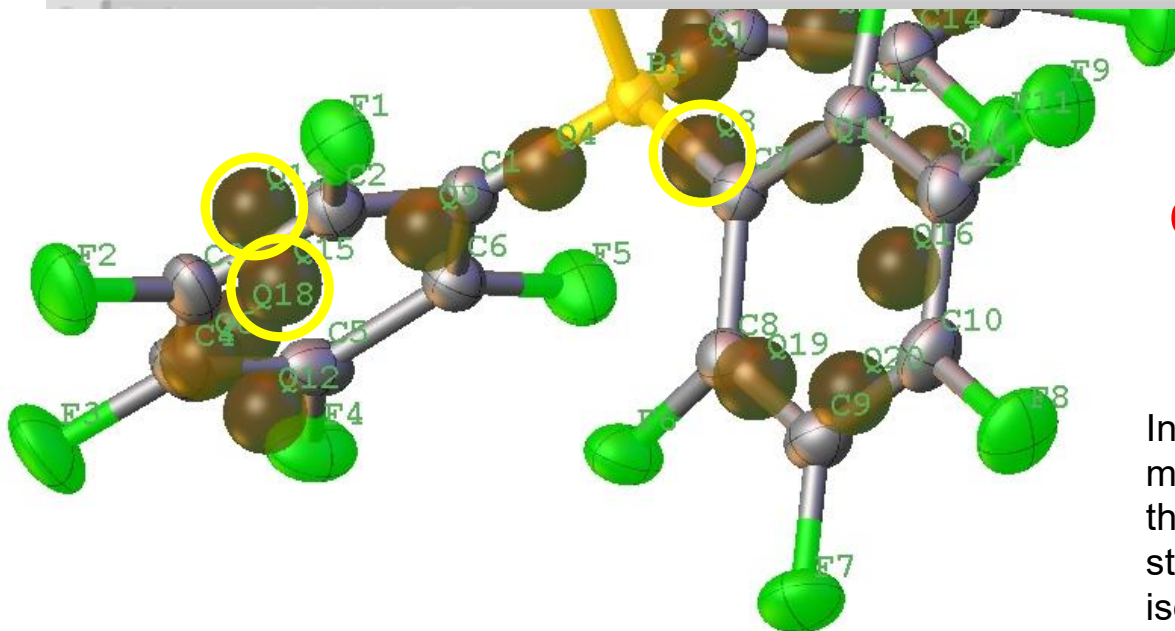
ent of hydrogen atoms, all
ity maxima are smaller than
can be ignored. ($0.5 \text{ e}^-/\text{\AA}^3$ is
the “allowable noise level.”)

Y	Z	Ueq	Peak
0.58	0.660	0.845	0.050
			0.280

- ~~Thermal movement of atoms~~
- ~~Hydrogen atoms~~
- Noise and artefacts

Observe the positions of the maxima, though! Is this really noise?

In a high-quality dataset, the remaining maxima correspond to electron density in the bonds. This is an artefact of our structural model: we assume a model of isolated, non-interacting, spherical atoms.



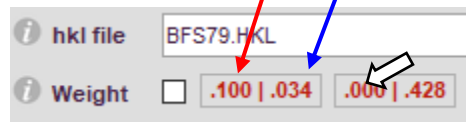
What happens behind the GUI ?

***.res:**

```
...  
WGHT 0.1000 ← current weighing scheme  
FVAR 0.3234
```

```
...  
HKLF 4  
END  
WGHT 0.0342 0.4326 ← proposed weighing scheme
```

The end: All lines
after END are ignored



Click on the numbers to copy
proposed to current weighing scheme

***.ins:**

```
...  
WGHT 0.0342 0.4326  
FVAR 0.3234  
...  
HKLF 4  
END  
WGHT 0.0342 0.4326
```



www.grammar.zone

Of course... we *could* do this by hand in the ins file
ourselves instead of using OLEX2. ☺

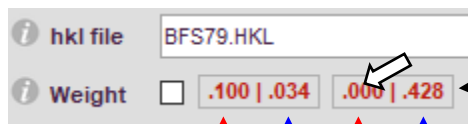
Refinement of the weighing scheme

*.lst:	<i>weak reflections</i>					<i>intense reflections</i>					
Fc/Fc(max)	0.000	0.012	0.023	0.034	0.047	0.061	0.079	0.101	0.135	0.201	1.000
Number in group	463	423	422	475	397	439	438	433	430	439	
GooF	1.406	1.493	1.414	1.317	1.076	1.029	0.813	0.734	0.633	0.453	
K	1.690	1.053	1.022	1.020	1.008	1.011	1.012	0.997	1.002	0.990	

We use an appropriate weighing scheme so that the GooF («Goodness of Fit») is independent from the intensity.

$$w = 1 / \sigma^2(F_o^2) + (aP)^2 + bP \quad P = 1/3(F_o^2 + 2F_c^2)$$

How do we refine the weighing scheme in OLEX2 :



red color indicates non-refined weighing scheme

current weighing scheme

proposed weighing scheme

***.res:**

...
WGHT **0.1000** ← current weighing scheme
FVAR 0.3234

...
HKLF 4

END
WGHT 0.0342 0.4326 ← proposed weighing scheme

The end: All lines after END are ignored

Refinement of the weighing scheme

*.lst:	<i>weak reflections</i>							<i>intense reflections</i>			
Fc/Fc(max)	0.000	0.012	0.023	0.034	0.047	0.061	0.079	0.101	0.135	0.201	1.000
Number in group	463	423	422	475	397	439	438	433	430	439	
GooF	1.406	1.493	1.414	1.317	1.076	1.029	0.813	0.734	0.633	0.453	
K	1.690	1.053	1.022	1.020	1.008	1.011	1.012	0.997	1.002	0.990	

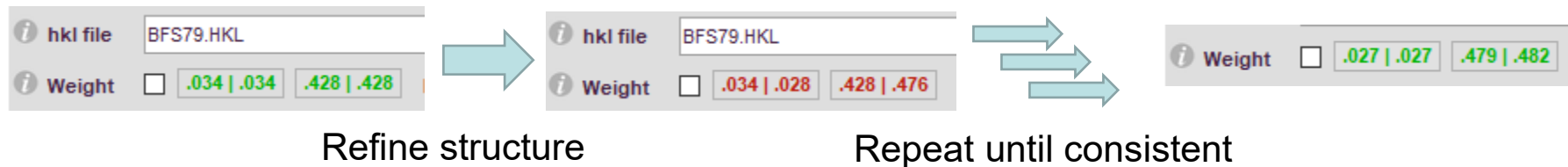
We use an appropriate weighing scheme so that the GooF («Goodness of Fit») is independent from the intensity.

$$w = 1 / \sigma^2(F_o^2) + (aP)^2 + bP \quad P = 1/3(F_o^2 + 2F_c^2)$$

How do we refine the weighing scheme in OLEX2 :

The screenshot shows the 'Weight' panel in OLEX2. The 'hkl file' is 'BFS79.HKL'. The 'Weight' section has two input fields: '.100 | .034' and '.000 | .428'. The first field is highlighted in red, and the second field is highlighted in blue. A red arrow points from the text 'current weighing scheme' to the red field. A blue arrow points from the text 'proposed weighing scheme' to the blue field. A black arrow points from the text 'red color indicates non-refined weighing scheme' to the red field.

Click on the numbers to copy proposed to current weighing scheme



Refinement of the weighing scheme

*.lst:	<i>weak reflections</i>					<i>intense reflections</i>					
Fc/Fc(max)	0.000	0.012	0.023	0.034	0.047	0.061	0.079	0.101	0.135	0.201	1.000
Number in group	463	423	422	475	397	439	438	433	430	439	
GooF	1.406	1.493	1.414	1.317	1.076	1.029	0.813	0.734	0.633	0.453	
K	1.690	1.053	1.022	1.020	1.008	1.011	1.012	0.997	1.002	0.990	

We use an appropriate weighing scheme so that the GooF («Goodness of Fit») is independent from the intensity.

$$w = 1 / \sigma^2(F_o^2) + (aP)^2 + bP \quad P = 1/3(F_o^2 + 2F_c^2)$$

hkl file: BFS79.HKL

Weight: ☐ .100 | .034 .000 | .428 EXTI ☐



hkl file: BFS79.HKL

Weight: ☒ .100 | .034 .000 | .428 EXTI ☐

Activate « Automatic Update » of the weighing scheme after each refinement. (Do that only after you are done with the hydrogen atoms).

Refine until consistent



hkl file: BFS79.HKL

Weight: ☒ .027 | .027 .479 | .482 EXTI ☐

*.lst:											
Fc/Fc(max)	0.000	0.012	0.023	0.035	0.047	0.062	0.079	0.101	0.135	0.202	1.000
Number in group	466.	418.	459.	438.	419.	417.	442.	435.	427.	438.	
GooF	1.117	1.023	1.008	1.059	0.974	1.102	0.990	1.064	1.056	0.993	
K	1.732	1.049	1.028	1.018	1.008	1.011	1.013	0.997	1.004	0.992	

Finishing the refinement

Make sure that the formula is correct



Cycle indicates
disagreement of UNIT
instruction and atom list

*.lst:

...

** Cell contents from UNIT instruction and atom list do not agree **

Unit-cell contents from UNIT instruction and atom list resp.

C	44.00	44.00
[...]		
O	1.00	0.00

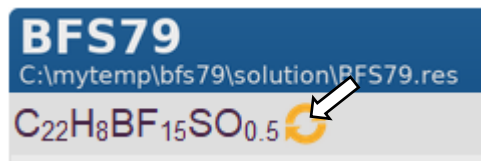
Attention: you **have to** verify if the error is in
the **model** or in the **UNIT** command!

Typical causes:

- Extra molecules found after unit instruction was given (solvent, etc.) : **UNIT is wrong**
- HADD added Hs incorrectly : **model is wrong** → do something
- Errors in disorder treatment (particularly if numbers are non integers) : **model is wrong** → do something

Only if UNIT is wrong:

Clicking on the cycle adapts UNIT command to atom list



*.res

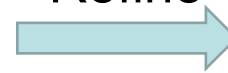
SFAC C H B F S O
UNIT 44 16 2 30 2 1



*.ins

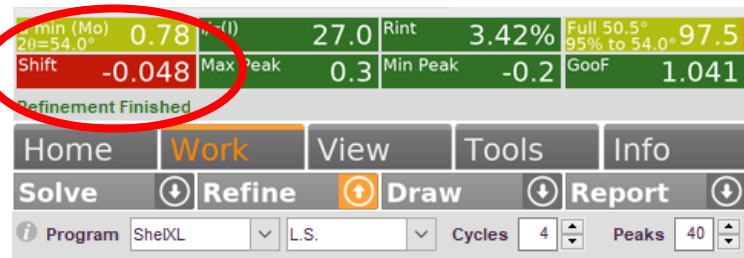
SFAC C H B F S
UNIT 44 16 2 30 2

Refine



Finishing the refinement

Let the refinement converge completely



.lst:

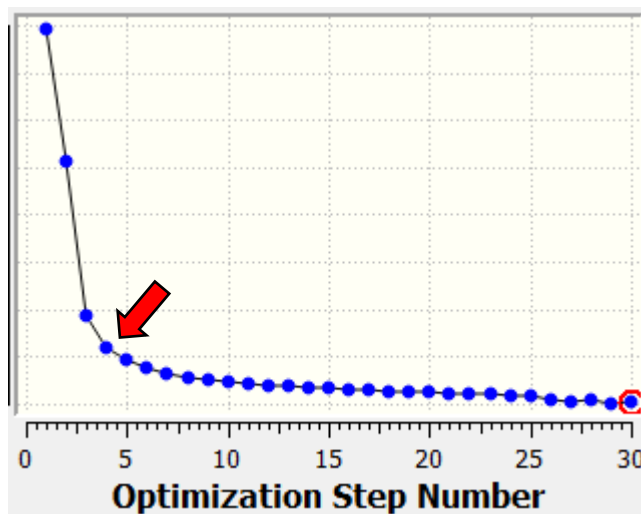
Least-squares cycle 4

wR2 = 0.0836 before cycle 2 for 4359 data and 352 / 352 parameters

[...]

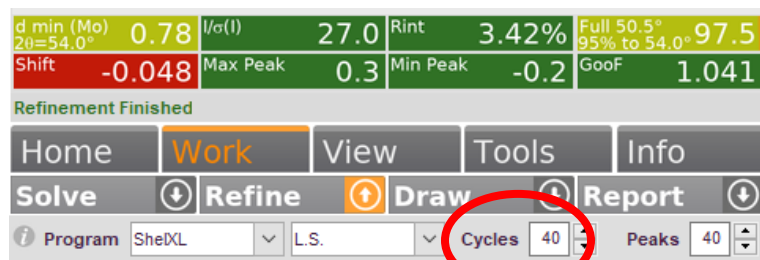
Mean shift/esd = 0.013 Maximum = -0.048 for y F13

Max. shift = 0.000 Å for C4 Max. dU = 0.000 for F2

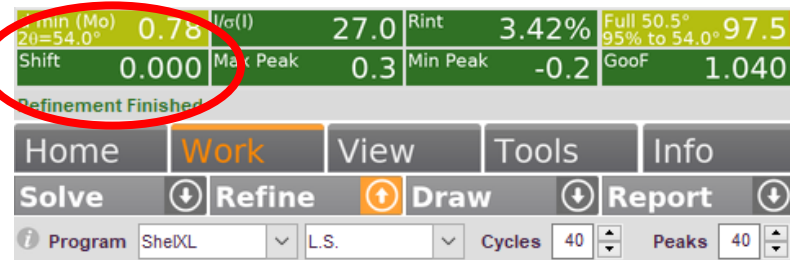


Once we are done with our changes to the model, there is simply no reason, other than laziness, not to let the refinement reach the minimum!

«max. shift» has to be smaller than $< 0.003 \text{ Å}$



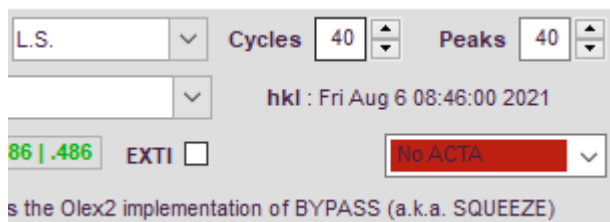
refine



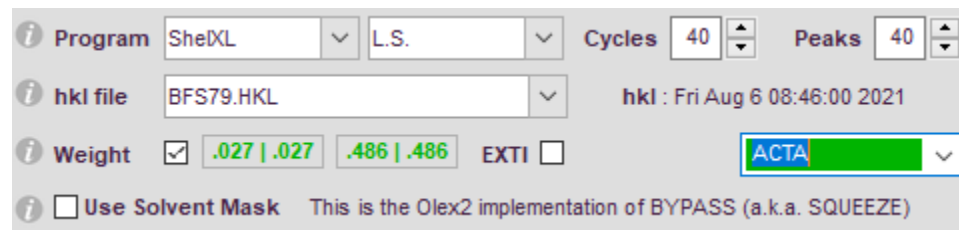
*.ins: L.S. 40

Finishing the refinement

Generate *.cif (and *.fcf) with the ACTA command



L.S. Cycles 40 Peaks 40
hkl : Fri Aug 6 08:46:00 2021
86 | .486 EXTI ☐ No ACTA
s the Olex2 implementation of BYPASS (a.k.a. SQUEEZE)



Program SheXL L.S. Cycles 40 Peaks 40
hkl file BFS79.HKL hkl : Fri Aug 6 08:46:00 2021
Weight ☒ .027 | .027 .486 | .486 EXTI ☐ ACTA
☐ Use Solvent Mask This is the Olex2 implementation of BYPASS (a.k.a. SQUEEZE)

*.ins:

```
...  
SFAC C H N F P  
UNIT 56 60 16 24 4  
L.S. 40  
FMAP 2  
...
```

When generating the CIF, ACTA imposes some “quality control”

- OMIT >0 is not permitted
- FMAP 2 is enforced
- LIST 4 is enforced

*.ins:

```
...  
SFAC C H N F P  
UNIT 56 60 16 24 4  
L.S. 40  
ACTA  
FMAP 2  
...
```

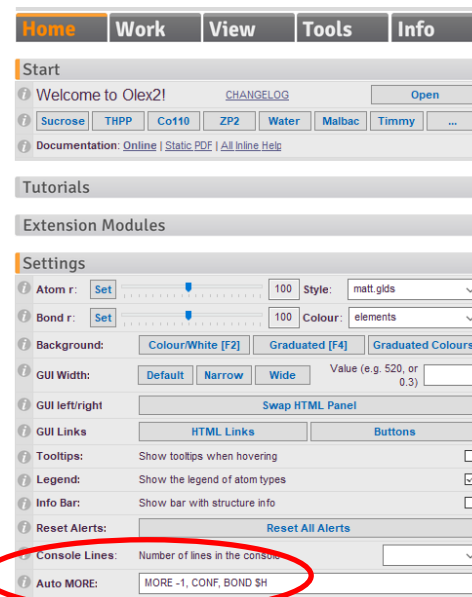
BOND \$H and CONF

*.ins

BOND \$H : Includes bond information (including H atoms in the CIF. Required by CHECKCIF.

CONF : Includes torsion angles with esd in CIF

OLEX2 always includes these commands automatically. This is governed in HOME – SETTINGS
(Do not change the MORE -1. OLEX2 needs this.)

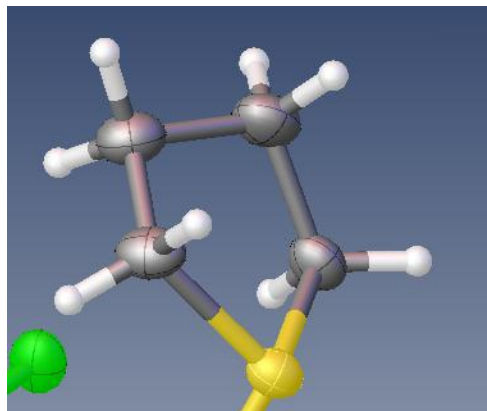


Home Work View Tools Info
Start
Welcome to Olex2! CHANGELOG Open
Sucrose THPP Co10 ZP2 Water Malbac Timmy ...
Documentation: Online Static PDF All Inline Help
Tutorials
Extension Modules
Settings
Atom r: Set 100 Style: matt.gids
Bond r: Set 100 Colour: elements
Background: Colour/White [F2] Graduated [F4] Graduated Colours
GUI Width: Default Narrow Wide Value (e.g. 520, or 0.3)
GUI left/right Swap HTML Panel
GUI Links HTML Links Buttons
Tooltips: Show tooltips when hovering
Legend: Show the legend of atom types
Info Bar: Show bar with structure info
Reset Alerts: Reset All Alerts
Console Lines: Number of lines in the console
Auto MORE: MORE -1, CONF, BOND \$H

Take-home message

Many/most standard refinement features can be (easier) addressed via the GUI. When should we bother opening the .res or .ins file?

- **Whenever** you find it more convenient (e. g. disorder refinements)
- **At least once** to double-check for anything unusual before making the final CIF: Unsorted atoms, etc.
- **Every time** when you run into problems



<chem>C22H8BF15S</chem>				      	
a = 9.5892(6)	α = 111.904(6)°	Z = 2	 12.3	R_1	3.13 %
b = 10.8264(10)	β = 96.814(7)°	Z' = 1			
c = 11.3959(9)	γ = 100.727(7)°	V = 1055.29(17)	wR_2	8.36 %	
d min (Mo)	0.78	l/σ(l)	27.0	Rint	3.42%
2θ = 54.0°				Full 50.5°	97.5
				95% to 54.0°	
Shift	0.000	Max Peak	0.3	Min Peak	-0.2
				GooF	1.041
Cell contents from UNIT instruction and atom list do not agree					

Checkcif:

PLAT041 Calc. and Reported SumFormula Strings Differ
Sum formula C22 **H7.95** B F15 S C22 H8 B F15 S

OLEX2 : edit res

[...]

AFIX 23

H19A 2 0.50169 0.10539 0.69766 **0.98700** -1.20000

H19B 2 0.62439 0.04681 0.74976 **0.95578** -1.20000

AFIX 0